



Electrical Data Sheet for Socket SP5 and SP6 Processors

Publication# **56897** Revision: **0.83**
Issue Date: **January 2025**



Socket SP5 和 SP6 处理器的 电气数据表

56897 083 出版编号# .修订版:
2025年1月 期号:

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Revision History

Date	Revision	Description
January 2025	0.83	Updated Chapter 3.
January 2024	0.82	Updated Figure 3 and Table 16.
October 2023	0.81	Added Section 1.1. Updated Table 16. Updated Table 17 notes. Added note in Chapter 16.
August 2023	0.80	Revised Table 1.
July 2023	0.78	Updated Table 1.
March 2023	0.77	Added I3C pin parameters in Table 10.
February 2023	0.76	Added support for Family 1Ah Models 00h–0Fh processors. Added Bidirectional pin parameters. Updated eSPI timing information. Updated Figure 11.
September 2022	0.75	Added support for Family 19h Models A0h–AFh processors. Updated JTAG timing, eSPI timing, reset sequencing, and power sequencing information.
April 2022	0.74	Added Table 7 and Table 11. Updated Chapter 3.
February 2022	0.73	Corrected t ₂ description in Table 21.
December 2021	0.72	Added support for Family 19h Models 10h–1Fh Socket SP6 processors.
November 2021	0.71	Updated eSPI output timing. Added GPIO maximum drive strength specification. Updated Bidirectional pin list.
August 2021	0.70	Updated I3C and Bidirectional pin lists. Added note regarding PCIe reference clock buffer.
April 2021	0.60	Updated I3C tables, reset sequencing, and power sequencing information.
February 2021	0.50	Initial NDA version.

版本历史

Date	版本	描述
2025年1月	0.83	更新第3章。
2024年1月	0.82	更新图3和表16。
2023年10月	0.81	添加了第1.1节。更新了表16。更新了表17的注释。在16章中添加了注释。
2023年8月	0.80	修订了表1。
2023年7月	0.78	更新了表1。
2023年3月	0.77	在表10中添加了I3C引脚参数。
2023年2月	0.76	添加了对Family 1Ah型号00h–0Fh处理器的支持。添加了双向引脚参数。更新了eSPI时序信息。更新了图11。
2022年9月	0.75	添加了对Family 19h型号A0h–AFh处理器的支持。更新了JTAG时序、eSPI时序、复位时序和电源时序信息。
2022年4月	0.74	添加了表7和表11。更新了第3章。
2022年2月	0.73	修正了表21中t2的描述。
2021年12月	0.72	添加了对Family 19h型号10h–1Fh插槽SP6处理器的支持。
2021年11月	0.71	更新了 eSPI 输出时序。添加了 GPIO 最大驱动强度规范。更新了双向引脚列表。
2021 年 8 月	0.70	更新了 I3C 和双向引脚列表。添加了关于 PCIe 参考时钟缓冲器的说明。
2021 年 4 月	0.60	更新了 I3C 表格、复位时序和电源时序信息。
2021 年 2 月	0.50	初始NDA版本。

Chapter 1 Introduction

This document contains power sequencing requirements and various I/O electrical requirements for the SP5 and SP6 infrastructures.

- Socket SP5
 - Family 19h Models 10h–1Fh
 - Family 19h Models A0h–AFh
 - Family 1Ah Models 00h–0Fh
 - Family 1Ah Models 10h–1Fh
- Socket SP6
 - Family 19h Models A0h–AFh

1.1 Reference Documents

For additional information, refer to the following AMD publications to supplement this document:

- *Infrastructure Roadmap (IRM)*
 - *Infrastructure Roadmap (IRM) for Socket SP5 Processors*, publication #56943
 - *Infrastructure Roadmap (IRM) for Socket SP6 Processors*, publication #57240
- *Functional Data Sheet (FDS)*
 - *Socket SP5 Processor Functional Data Sheet*, publication #56900
 - *Socket SP6 Processor Functional Data Sheet*, publication #57388
- *Motherboard Design Guide (MBDG)*
 - *Socket SP5 Processor Motherboard Design Guide*, publication #56870
 - *Socket SP6 Processor Motherboard Design Guide*, publication #57387
- *Processor Programming Reference (PPR)*
 - *Processor Programming Reference (PPR) for AMD Family 19h Models 10h–1Fh Processors*, publication #55901
 - *Processor Programming Reference (PPR) for AMD Family 19h Models A0h–AFh Processors*, publication #57228
 - *Processor Programming Reference (PPR) for AMD Family 1Ah Models 00h–0Fh Processors*, publication #57238
 - *Processor Programming Reference (PPR) for AMD Family 1Ah Models 10h–1Fh Processors*, publication #57883
- *Memory Population Guidelines*

第一章 简介

本文档包含 SP5 和 SP6 基础架构的电源时序要求和各种 I/O 电气要求。

- Socket SP5
 - 家族19h模型10h–1Fh
 - 家族19h模型A0h–AFh
 - 家族1Ah型号00h–0Fh
 - 家族 1Ah 模型 10h–1Fh
- Socket SP6
 - Family 19h 型号 A0h–AFh

1.1 参考文档

如需更多信息，请参考以下AMD出版物以补充本文件：

- 基础设施路线图 (*IRM*)
 - *Socket SP5 处理器的基础设施路线图 (IRM)*，出版物编号 #56943
 - *Socket SP6 处理器的基础设施路线图 (IRM)*，出版物编号 #57240
- 功能数据表 (*FDS*)
 - *Socket SP5 处理器功能数据表*，发布编号 #56900
 - *Socket SP6 处理器功能数据表*，发布编号 #57388
- 主板设计指南 (*MBDG*)
 - *Socket SP5 处理器主板设计指南*，发布编号 #56870
 - *Socket SP6 处理器主板设计指南*，发布编号 #57387
- 处理器编程参考 (*PPR*)
 - *AMD 家族 19h 型号 10h–1Fh 处理器的处理器编程参考 (PPR)*，出版物编号 #55901
 - *AMD 家族 19h 型号 A0h–AFh 处理器的处理器编程参考 (PPR)*，出版物编号 #57228
 - *AMD 家族 1Ah 型号 00h–0Fh 处理器的处理器编程参考 (PPR)*，出版物编号 #57238
 - *AMD 家族 1Ah 型号 10h–1Fh 处理器的处理器编程参考 (PPR)*，出版物编号 #57883
- 内存填充指南

- Memory Population Guidelines for AMD Family 19h Models 10h–1Fh and Models A0h–AFh Socket SP5 Processors, publication #56960
- Memory Population Guidelines for AMD Family 19h Models A0h–AFh Socket SP6 Processors, publication #57609
- Memory Population Guidelines for AMD Family 1Ah Models 00h–0Fh and Models 10h–1Fh Socket SP5 Processors, publication #58266
- AMD Serial VID Interface Version 3 (SVI3) Specification, publication #56413

- AMD Family 19h 型号 10h–1Fh 和型号 A0h–AFh Socket SP5 处理器的内存填充指南，出版物编号 #56960
- AMD Family 19h 型号 A0h–AFh Socket SP6 处理器的内存填充指南，出版物编号 #57609
- AMD Family 1Ah 型号 00h–0Fh 和型号 10h–1Fh Socket SP5 处理器的内存填充指南，出版物编号 #58266
- AMD 序列 VID 接口版本 3 (SVI3) 规范，发布编号 #56413

Chapter 2 Absolute Maximum Ratings

Table 1 specifies absolute maximum ratings. Absolute maximum ratings lie outside the operating limits of the processor. Long-term reliability and functionality are expected within operating conditions. Systems using this device must be designed to ensure that absolute maximum ratings are not violated.

At conditions outside the operating limits of the processor, but within the absolute maximum ratings, functionality and long-term reliability of the device are not guaranteed or expected. If the device is returned to within the operating conditions, the device may function, but the long-term reliability may be degraded.

Device functionality and reliability will be compromised when a device is operated outside the absolute maximum ratings. Furthermore, when a device is returned to within operating conditions the device may not work and its long-term reliability will have been degraded.

Table 1. Absolute Maximum Ratings

Characteristic	Range
VDDCR_CPU0 and VDDCR_CPU1 supply voltages relative to VSS	−0.3 to 1.70 V
VDDCR_SOC supply voltage relative to VSS	−0.3 to 1.35 V
VDDIO supply voltage relative to VSS	−0.3 to 1.28 V
VDDIO_AUDIO supply voltage relative to VSS	−0.3 to 1.98 V
VDDBT_RTC_G supply voltage relative to VSS	−0.3 to 3.30 V
VDD_11_S3 supply voltage relative to VSS	−0.3 to 1.20 V
VDD_18_S5 supply voltage relative to VSS	−0.3 to 1.98 V
VDD_33_S5 supply voltage relative to VSS	−0.3 to 3.63 V

第 2 章 绝对最大额定值

表 1 指定绝对最大额定值。绝对最大额定值超出处理器的运行限制。预期在运行条件下长期可靠性和功能。使用此设备的系统必须设计为确保不违反绝对最大额定值。

在超出处理器工作范围的条件下，但在绝对最大额定值范围内，设备的性能和长期可靠性无法保证或预期。如果设备恢复到工作条件范围内，设备可能能够正常工作，但长期可靠性可能会下降。

当设备在绝对最大额定值之外运行时，其功能性和可靠性将受到损害。此外，当设备恢复到工作条件时，它可能无法正常工作，并且其长期可靠性可能会下降。

表 1. 绝对最大额定值

特性	范围
VDDCR_CPU0 和 VDDCR_CPU1 电源电压相对于 VSS	−0.3 至 1.70 V
VDDCR_相对于VSS的SOC电源电压	−0.3至1.35 V
VDDIO相对于VSS的电源电压	−0.3至1.28 V
VDDIO_AUDIO相对于VSS的电源电压	−0.3至1.98 V
VDDBT_RTC_G 相对于 VSS 的供电电压	−0.3 至 3.30 V
VDD_11_S3 相对于 VSS 的供电电压	−0.3 至 1.20 V
VDD_18_S5 相对于 VSS 的供电电压	−0.3 至 1.98 V
VDD_33_S5 供电电压相对于 VSS	−0.3 至 3.63 V

Chapter 3 DDR SDRAM Interface

Silicon is DDR5 JEDEC compliant.

Socket SP5 supports transfer rates from 3600 to 6400 MT/s, and Socket SP6 supports transfer rates from 3600 to 4800 MT/s. For more information, refer to the *Memory Population Guidelines* documents.

第3章 DDR SDRAM 接口

硅是符合 DDR5 JEDEC 标准的。

Socket SP5 支持 3600 至 6400 MT/s 的传输速率，而 Socket SP6 支持 3600 至 4800 MT/s 的传输速率。有关更多信息，请参阅内存配置指南文档。

Chapter 4 PCI Express® Interface

Refer to the *PCI Express Base Specification Revision 5.0, Version 1.0*. Any deviation from the specifications is listed in this document.

第 4 章 PCI Express® 接口

参考 PCI Express 基础规范修订版 5.0，版本 1.0。任何与规范不符的内容均列在此文档中。

Chapter 5 xGMI Interface

The xGMI receivers support PCIe Gen5 compatible channels.

Contact your local AMD representative for simulation models.

第 5 章 xGMI 接口

xGMI 接收器支持 PCIe Gen5 兼容通道。

联系您当地的 AMD 代表以获取仿真模型。

Chapter 6 Clock Pins

For additional information on common mode, refer to the *PCI Express Base Specification*.

Table 2. Combined AC and DC Operating Conditions for GPP_CLK01N, GPP_CLK01P, GPP_CLK02N, GPP_CLK02P, GPP_CLK03N, GPP_CLK03P, GPP_CLK04N, GPP_CLK04P, GPP_CLK05N, GPP_CLK05P, REFCLK100SSC_N/GPP_CLK10N, REFCLK100SSC_P/GPP_CLK10P, GPP_CLK11N, GPP_CLK11P, GPP_CLK12N, GPP_CLK12P, GPP_CLK13N, GPP_CLK13P, GPP_CLK14N, GPP_CLK14P, GPP_CLK15N, and GPP_CLK15P

Symbol	Parameter	Min	Max	Unit
freq	Frequency (output)	48	100	MHz
t _{Fall/Rise}	Measured from -150 mV to +150 mV	0.6	4.0	V/ns
V _{IH}	Differential input high voltage	150	—	mV
V _{IL}	Differential input low voltage	—	-150	
V _{Cross}	Absolute crossing point voltage	250	550	
V _{Crossdelta}	Variation of V _{Cross} over all rising clock edges	—	140	
—	Duty cycle	40	60	%
—	Output driver termination	40	60	Ω

- Notes:
- External input clock is supported on GPP_CLK02N, GPP_CLK02P, GPP_CLK12N, GPP_CLK12P.
 - When a buffer is added to fan out the PCIe reference clock, choose buffers that contribute a minimum amount of jitter. AMD recommends a fan out buffer with a jitter of less than 20 fSec.

第6章时钟引脚

有关常用模式的更多信息，请参阅PCI Express基本规范。

表2. GPP_CLK01N、GPP_CLK01P、GPP_CLK02N、GPP_CLK02P、GPP_CLK03N、GPP_CLK03P、GPP_CLK04N、GPP_CLK04P、GPP_CLK05N、GPP_CLK05P、REFCLK100SSC_N/GPP_CLK10N、REFCLK100SSC_P/GPP_CLK10P、GPP_CLK11N、GPP_CLK11P、GPP_CLK12N、GPP_CLK12P、GPP_CLK13N、GPP_CLK13P、GPP_CLK14N和GPP_CLK14P的联合交流和直流工作条件

符号	参数	Min	Max	Unit
freq	频率（输出）	48	100	MHz
t _{Fall/Rise}	从 -150 mV 测量到 +150 mV	0.6	4.0	V/ns
V _{IH}	差分输入高电压	150	—	mV
V _{IL}	差分输入低电压	—	-150	
V _{Cross}	绝对交叉点电压	250	550	
V _{Crossdelta}	VCross在所有上升时钟边沿上的变化	—	140	
—	占空比	40	60	%
—	输出驱动端接	40	60	Ω

- 备注:
- GPP_CLK02N、GPP_CLK02P、GPP_CLK12N、GPP_CLK12P 支持外部输入时钟。
 - 当添加缓冲器以扇出 PCIe 参考时钟时，选择抖动最小的缓冲器。AMD 建议使用抖动小于 20 fSec 的扇出缓冲器。

Table 3. Combined AC and DC Operating Conditions for X48M_X1 and X48M_X2

Symbol	Parameter	Min	Typical	Max	Unit	Notes
f _{xoclk_48}	Crystal or external clock generator reference frequency	—	48	—	MHz	3, 4
f _{acc}	External clock generator output frequency accuracy	-50	—	50	PPM	1, 4, 7
C _{In}	Input capacitance	8.4	—	8.4	pF	
D _L	Drive level	—	200	600	μW	4
ESR	Equivalent series resistance	—	10	50	Ω	4
V _{IH}	Input high level	1.62	1.8	—	V	5
V _{IL}	Input low level	—	0	0.18		5
DC	Duty cycle	45	50	55	%	5
t _{IR}	48 MHz rise slew rate	1	—	—	V/ns	5, 6
t _{IF}	48 MHz fall slew rate	1	—	—		5, 6
t _{Short}	Short term peak-to-peak jitter	—	—	200	ps	5
t _{IJLT}	48 MHz long term jitter requirement (10 μs after scope trigger)	—	—	250		5
V _{i_low_PAD_XTALIN}	Voltage level of logic LOW of crystal/oscillator signal PAD_XTALIN. Referenced to VSS_XTAL	—	0	—	V	4
V _{i_high_PAD_XTAQLI}	Voltage level of logic HIGH of crystal/oscillator	1.62	—	1.98		4

- Notes:
- 25°C ± 3°C
 - 0–85°C
 - Maximum 30 PPM
 - Crystal mode
 - External oscillator mode
 - For acceptable duty-cycle performance, t_{IR} and t_{IF} should be matched to within 10% of each other
 - This is the maximum PPM tolerated by the processor; but other factors such as time-keeping accuracy within the overall system may require a smaller PPM. Refer to the “Socket SP5 Processor Motherboard Design Guide”, publication #56870, for suggested PPM values.

表 3. X48M_X1 和 X48M_X2 的交流电和直流电工作条件

符号	参数	Min	典型	Max	Unit	备注
fxoclk_48_	晶体或外部时钟发生器参考频率	—	48	—	MHz	3, 4
f _{acc}	外部时钟发生器输出频率精度	-50	—	50	PPM	1, 4, 7
C _{In}	输入电容	8.4	—	8.4	pF	
D _L	驱动电平	—	200	600	μW	4
ESR	等效串联电阻	—	10	50	Ω	4
V _{IH}	输入高级	1.62	1.8	—	V	5
V _{IL}	输入低级	—	0	0.18		5
DC	占空比	45	50	55	%	5
t _{IR}	48 MHz 上升速率	1	—	—	V/ns	5, 6
t _{IF}	48 MHz 下降速率	1	—	—		5, 6
t _{Short}	短期峰峰值抖动	—	—	200	ps	5
t _{IJLT}	48 MHz 长期抖动要求 (10 μs 扫描触发后)	—	—	250		5
V _{i_低_PAD_XTALIN}	逻辑低电平电压等级的晶体/振荡器信号PAD_XTALIN. 相对于VSS XTAL_	—	0	—	V	4
V _{i_高_PAD_XTAQLI}	逻辑高电平电压等级的晶体/振荡器	1.62	—	1.98		4

- 备注:
- 25°C ± 3°C
 - 0–85°C
 - 最大 30 PPM
 - 晶体模式
 - 外部振荡器模式
 - 为了达到可接受的占空比性能，t_{IR} 和 t_{IF} 应该彼此匹配在 10% 以内
 - 这是处理器可容忍的最大PPM，但其他因素（如整个系统内的时间保持精度）可能需要更小的PPM。有关建议的PPM值，请参阅“Socket SP5处理器主板设计指南”，出版物编号#56870。

Table 4. Combined AC and DC Operating Conditions for X32K_X1 and X32K_X2

Symbol	Parameter	Min	Typical	Max	Unit	Notes
f _{xoclk_32}	Crystal reference frequency	—	32.768	—	KHz	
f _{acc}	External clock generator output frequency accuracy	-100	—	100	PPM	1, 2
C _{In}	Input capacitance	9.0	—	9.5	pF	5
V _{IH}	Input high	0.8× V _{DDBT_RTC_G}	—	—	V	3, 4
V _{IL}	Input low	—	—	0.2× V _{DDBT_RTC_G}		3, 4
t _{IR}	32 kHz rise slew rate	0.066	—	—	V/ns	3, 4
t _{IF}	32 kHz fall slew rate	0.066	—	—		3, 4
t _{IJLT}	32 kHz long term jitter requirement	—	—	100	ns	3, 4
t _{Short}	Short term peak-to-peak jitter	—	—	100		3, 4
—	Duty cycle	48	50	52	%	3, 4

- Notes:
- Crystal mode
 - This is the maximum PPM tolerated by the processor, but other factors such as time-keeping accuracy within the overall system may require a smaller PPM. Refer to the “Socket SP5 Processor Motherboard Design Guide”, publication #56870, for suggested PPM values.
 - $V_{DDBT_RTC_G} = 1.5$ or 1.8 V
 - In external oscillator mode:
If the CLK from RTC IC is designed as G3 domain, you can connect it to X32K_X2 to save battery power.
If the CLK from RTC IC is designed as G3 domain, you can connect it to X32K_X1 to avoid battery power leakage to the pull high resistor.
 - In Crystal mode, C_{In} could be adjusted to get least PPM error.

Table 5. Combined AC and DC Operating Conditions for REFCLK100SSC_N and REFCLK100SSC_P

Symbol	Parameter	Min	Max	Unit
freq	Frequency (input and output)	48	100	MHz
t _{Fall/Rise}	Measured from -150 mV to +150 mV	0.6	4.0	V/ns
V _{IH}	Differential input high voltage	150	—	mV
V _{IL}	Differential input low voltage	—	-150	
V _{Cross}	Absolute crossing point voltage	250	550	
V _{Crossdelta}	Variation of V _{Cross} over all rising clock edges	—	140	
—	Duty cycle	40	60	%
—	Output termination	40	60	Ω

表 4. X32K_X1 和 X32K_X2 的交流及直流工作条件组合

符号	参数	Min	典型	Max	Unit	备注
fxoclk_32_	晶体参考频率	—	32.768	—	KHz	
f _{acc}	外部时钟发生器输出频率精度	-100	—	100	PPM	1, 2
C _{In}	输入电容	9.0	—	9.5	pF	5
V _{IH}	输入高	0.8× V _{DDBT_RTC_G_}	—	—	V	3, 4
V _{IL}	输入低	—	—	0.2× V _{DDBT_RTC_G_}		3, 4
t _{IR}	32 kHz 上升速率	0.066	—	—	V/ns	3, 4
t _{IF}	32 kHz 下降速率	0.066	—	—		3, 4
t _{IJLT}	32 kHz 长期抖动要求	—	—	100	ns	3, 4
t _{Short}	短期峰峰值抖动	—	—	100		3, 4
—	占空比	48	50	52	%	3, 4

- 备注:
- 晶体模式
 - 这是处理器能容忍的最大 PPM，但其他因素，如整个系统内的时间保持精度，可能需要更小的 PPM。请参考“Socket SP5 处理器主板设计指南”，出版物编号 #56870，以获取建议的 PPM 值。
 - $V_{DDBT_RTCG} = 1.5$ 或 1.8 V
 - 在外部振荡器模式下：
如果RTC IC的CLK设计为G3域，您可以将其连接到X32K_X2以节省电池电量。如果RTC IC的CLK设计为G3域，您可以将其连接到X32K_X1以避免电池电量泄漏到上拉电阻。
 - 在水晶模式下，C_{In} 可以调整以获得最小的 PPM 误差。

表5. REFCLK100SSC_N和REFCLK100SSC的联合交流和直流工作条件

符号	参数	Min	Max	Unit
freq	频率（输入和输出）	48	100	MHz
t _{Fall/Rise}	从 -150 mV 测量到 +150 mV	0.6	4.0	V/ns
V _{IH}	差分输入高电压	150	—	mV
V _{IL}	差分输入低电压	—	-150	
V _{Cross}	绝对交叉点电压	250	550	
V _{Crossdelta}	VCross在所有上升时钟边沿的变化	—	140	
—	占空比	40	60	%
—	输出端接	40	60	Ω

Chapter 7 Miscellaneous Pins

For input and bidirectional pins:

- Maximum V_{IH-DC} is the VDD voltage.
- Maximum V_{IH-AC} is 110% of the VDD voltage; overshoot is limited in duration to 25% of the clock period of the highest frequency of the pin's clock.
- Minimum V_{IL-DC} is ground.
- Minimum V_{IL-AC} is 10% of the VDD voltage below ground; undershoot is limited in duration to 25% of the clock period of the highest frequency of the pin's clock.
- For GPIO, VDD_18, VDD_18_S5 = 1.8 V.
- GPIO maximum drive strength is 8.34 mA.

7.1 Miscellaneous Input Pins

Table 6. Combined AC and DC Operating Conditions for RSMRST_L, TCK, TDI, TMS, TRST_L, DBREQ_L, FORCE_SELFREFRESH, PWR_GOOD, PROCHOT_L, PWR_BTN_L/AGPIO0, PCC0_L, and PCC1_L

Symbol	Parameter	Min	Max	Unit	Notes
V _{In}	Max input voltage	—	1.80	V	
V _{IH-DC 1.8V RX}	Input high voltage (logic 1)	1.20	1.80		
V _{IH-AC 1.8V RX}			1.98		1
V _{IL-DC 1.8V RX}	Input low voltage (logic 0)	0	0.70		
V _{IL-AC 1.8V RX}		-0.18			1
I _L	Input leakage current	—	5	μA	
C _{In}	Input capacitance	—	1.5	pF	
t _{PROCHOT_L}	PROCHOT_L assertion time	5	—	ms	

Notes:
1. Input overshoot and undershoot are limited in duration to 25% of the clock period of the highest frequency of the pin's clock.

第 7 章 其他引脚

对于输入和双向引脚：

- 最大 V_{IH-DC} 是 VDD 电压。
- 最大 V_{IH-AC} 为 VDD 电压的 110%；超调持续时间限制为最高频率引脚时钟周期的 25%。
- 最小 V_{IL-DC} 为地。
- 最小 V_{IL-AC} 为低于地 VDD 电压的 10%；下冲持续时间限制为最高频率引脚时钟周期的 25%。
- 对于 GPI O, VDD_18, VDD_18_S5 = 1.8 V。
- GPIO 最大驱动电流为 8.34 mA。

7.1 其他输入引脚

表 6. RSMRST_L、TCK、TDI、TMS、TRST_L、DBREQ_L、FORCE_SELFREFRESH、PWR_GOOD、PROCHOT_L、PWR_BTN_L/AGPIO0、PCC0_L 和 PCC1_L 的交流及直流工作条件

符号	参数	Min	Max	Unit	注释
V _{In}	最大输入电压	—	1.80	V	
VIH-DC 1.8V RX	输入高电压（逻辑1）	1.20	1.80		
VIH-AC 1.8V RX			1.98		1
VIL-DC 1.8V RX	输入低电压（逻辑0）	0	0.70		
VIL-AC 1.8V RX		-0.18			1
I _L	输入漏电流	—	5	μA	
C _{In}	输入电容	—	1.5	pF	
tPROCHOT _L	PROCHOT _L建立时间	5	—	ms	

备注：
1. 输入过冲和欠冲在持续时间上限制为引脚时钟最高频率的25%。

Table 7. Combined AC and DC Operating Conditions for RTC_LDO_SELECT

Symbol	Parameter	Min	Max	Unit
V _{In}	Max input voltage	—	3.3	V
V _{IH-DC}	Input high voltage (logic 1)	2.2	3.3	
V _{IH-AC}			3.46	
V _{IL-DC}	Input low voltage (logic 0)	0	0.7	
V _{IL-AC}		-0.3		
I _L	Input leakage current	—	5	uA
C _{In}	Input capacitance	—	1.5	pF

For TCK, TDI, and TMS timing, see Chapter 17.

7.2 Miscellaneous Output Pins

Table 8. Combined AC and DC Operating Conditions for TDO, SLP_S3_L, SLP_S5_L, AGPIO7, SMERR_L/AGPIO24, and NV_SAVE_L

Symbol	Parameter	Min	Max	Unit
V _{OH-DC}	DC output high voltage (logic 1) @ I = 2 mA	VDD_18_S5 – 0.45	—	V
V _{OL-DC}	DC output low voltage (logic 0) @ I = –2 mA	—	0.45	

For TDO timing, see Chapter 17.

表 7. RTC_LDO_SELECT 的联合交流和直流工作条件

符号	参数	Min	Max	Unit
V _{In}	最大输入电压	—	3.3	V
V _{IH-DC}	输入高电压（逻辑1）	2.2	3.3	
V _{IH-AC}			3.46	
V _{IL-DC}	输入低电压（逻辑0）	0	0.7	
V _{IL-AC}		-0.3		
I _L	输入漏电流	—	5	uA
C _{In}	输入电容	—	1.5	pF

关于TCK、TDI和TMS的定时，请参阅第17章。

7.2 其他输出引脚

表8. TDO、SLP_S3_L、SLP_S5_L、AGPIO7、SMERR_L/AGPIO24，以及NV_SAVE_L的联合交流和直流工作条件

符号	参数	Min	Max	Unit
VOH-DC	直流输出高电压（逻辑1） @ I = 2 mA	VDD 18 S5 – 0.45_ _	—	V
VOL-DC	DC输出低电压（逻辑0） @ I = –2 mA	—	0.45	

关于TDO时序，请参阅第17章。

7.3 Miscellaneous Bidirectional Pins

Table 9. Combined AC and DC Operating Conditions for I²C/AGPIO Pins:
I2C4_SCL/HP_SCL/UBM_SCL/AGPIO13,
I2C4_SDA/HP_SDA/UBM_SDA/AGPIO14,
I2C5_SCL/BMC_SCL/SMBUS1_SCL/AGPIO19, and
I2C5_SDA/BMC_SDA/SMBUS1_SDA/AGPIO20

Symbol	Parameter	Min	Max	Unit	Notes
V _{OH-DC} 1.8V Mode	DC output high voltage (logic 1) @ I = 2 mA	—	—	V	
V _{OL-DC} 1.8V Mode	DC output low voltage (logic 0) @ I = −2 mA	—	0.36		
V _{IH-DC} 1.8V Mode	Input high voltage (logic 1)	1.26	1.80		
V _{IH-AC} 1.8V Mode			1.98		1
V _{IL-DC} 1.8V Mode	Input low voltage (logic 0)	0	0.54		
V _{IL-AC} 1.8V Mode		-0.18			1
I _L 1.8V Mode	Input leakage current	—	10	μA	
C _{In} 1.8V Mode	Input capacitance	—	10	pF	

Notes:
1. Input overshoot and undershoot are limited in duration to 25% of the clock period of the highest frequency of the pin's clock.

The I²C interface can function in 1.8 V mode. For further details, refer to the *Functional Data Sheet (FDS)* and *Motherboard Design Guide (MBDG)* documents.

7.3 其他双向引脚

表 9. I²C/AGPIO 引脚的联合交流和直流工作条件：I2C4_SCL/HP_

SCL/UBM_SCL/AGPIO13, I2C4_SDA/HP_SDA/UBM_SDA/AGPIO14, I2

C5_SCL/BMC_SCL/SMBUS1_SCL/AGPIO19, 以及 I2C5 SDA/BMC

SDA/SMBUS1 SDA/AGPIO20_

—

—

符号	参数	Min	Max	Unit	备注
VOH-DC 1.8V 模式	直流输出高电压（逻辑 1） @ I = 2 mA	—	—	V	
VOL-DC 1.8V 模式	DC 输出低电压（逻辑 0） @ I = -2 mA	—	0.36		
VIH-DC 1.8V 模式	输入高电压（逻辑 1）	1.26	1.80		
VIH-AC 1.8V 模式			1.98		1
VIL-DC 1.8V 模式	输入低电压（逻辑 0）	0	0.54		
VIL-AC 1.8V 模式		-0.18			1
IL 1.8V 模式	输入漏电流	—	10	μA	
CIn 1.8V 模式	输入电容	—	10	pF	

备注：

1. 输入过冲和欠冲在持续时间上限制为引脚时钟最高频率的25%。

I²C 接口可以在 1.8 V 模式下工作。有关详细信息，请参阅功能数据表 (FDS) 和主板设计指南 (MBDG) 文档。

Table 10. Combined AC and DC Operating Conditions for I³C Pins:
I3C0_SCL/I2C0_SCL/SPD0_SCL/SMBUS0_SCL/AGPIO145,
I3C0_SDA/I2C0_SDA/SPD0_SDA/SMBUS0_SDA/AGPIO146,
I3C1_SCL/I2C1_SCL/SPD1_SCL/AGPIO147, I3C1_SDA/I2C1_SDA/SPD1_SDA/AGPIO148,
I3C2_SCL/I2C2_SCL/SPD2_SCL/AGPIO149, I3C2_SDA/I2C2_SDA/SPD2_SDA/AGPIO150,
I3C3_SCL/I2C3_SCL/SPD3_SCL/AGPIO151, and
I3C3_SDA/I2C3_SDA/SPD3_SDA/AGPIO152

Symbol	Parameter	Min	Max	Unit	Notes
V _{OH} 1.1V Mode	Output high voltage (logic 1) @ I = 4 mA	0.75×VDD_11_S3	—	V	
V _{OL} 1.1V Mode	Output low voltage (logic 0) @ I = 4 mA	—	0.25×VDD_11_S3		
V _{IH-DC} 1.1V Mode	Minimum DC voltage at PAD pin that will produce a logic high output on 1.1V receiver	0.76	—		
V _{IH-AC} 1.1V Mode		0.92	—		
V _{IL-DC} 1.1V Mode	Maximum DC voltage at PAD pin that will produce logic low output on 1.1V receiver	—	0.32		
V _{IL-AC} 1.1V Mode		—	0.095		
I _L 1.1V Mode	Input leakage current (external pull-up of 3 kΩ) VDD = VDD_11_S3	—	5	uA	
C _{In} 1.1V Mode	Input capacitance VDD = VDD_11_S3	—	5	pF	
V _{HYS}	Input hysteresis	100	—	mV	
V _{IH_Min}	Input high voltage	0.7×VDD_11_S3	—	V	
V _{IL_Max}	Input low voltage	—	0.3×VDD_11_S3		
V _{SLPRV}	Receive buffer slope reversal	150	—	mV _{PK-PK}	1

Notes:

1. Slope reversal that occurs between V_{IH_Min} and V_{IL_Max} on either the rising or the falling edge, must be less than V_{SLPRV} (see Figure 1a). Rising edge slope reversal, where the signal first rises above V_{IH_Min} and then falls below V_{IH_Min}, must not drop more than V_{SLPRV} below V_{IH_Min} (see Figure 1b). Falling edge slope reversal, where the signal first drops below V_{IL_Max} then rises above V_{IL_Max}, must not rise more than V_{SLPRV} above V_{IH_Min} (see Figure 1c).

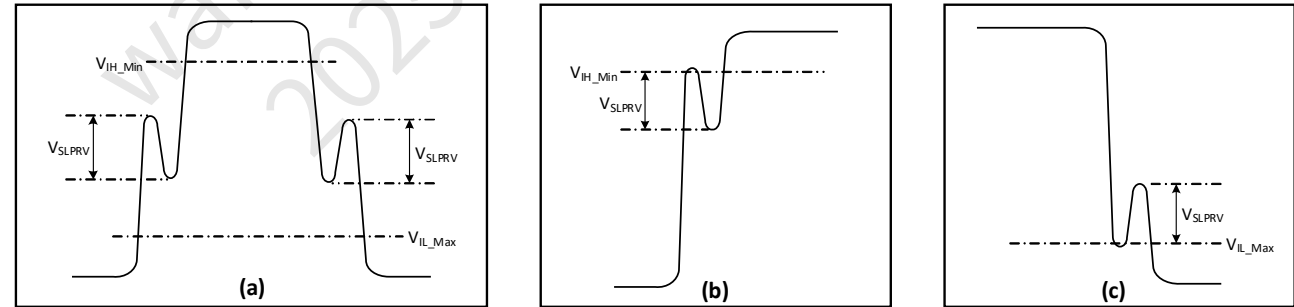


Figure 1. Slope Reversal Specification

表 10. 结合交流和直流 O peratin g I³C引脚的条件:
I3C0_SCL/I2C0_SCL/SPD0_SCL/SMBUS0_SCL/AGPIO145 ,
I3C0_SDA/I2C0_SDA/SPD0_SDA/SMBUS0_SDA/AGPIO146 ,
I3C1_SCL/I2C1_SCL/SPD1_SCL/AGPIO147, I3C1_SDA/I2C1_SDA/SPD1_SDA/AGPIO148,
I3C2_SCL/I2C2_SCL/SPD2_SCL/AGPIO149, I3C2_SDA/I2C2_SDA/SPD2_SDA/AGPIO150,
I3C3_SCL/I2C3_SCL/SPD3_SCL/AGPIO151, 以及I3C3
SDA/I2C3_SDA/SPD3_SDA/AGPIO152_ _ _

符号	参数	Min	Max	Unit	备注
VOH 1.1V 模式	输出高电压（逻辑 1） @ I = 4 mA	0.75×VDD 11 S3_ _	—	V	
VOL 1.1V 模式	输出低电压（逻辑 0） @ I = 4 mA	—	0.25×VDD 11 S3_ _		
VIH-DC 1.1V 模式	PAD 的最小直流电压 产生逻辑的引脚 1.1V 接收器的高输出	0.76	—		
VIH-AC 1.1V 模式		0.92	—		
VIL-DC 1.1V 模式	PAD 的最大直流电压 产生逻辑低的引脚 在1.1V接收器上输出	—	0.32		
VIL-AC 1.1V模式		—	0.095		
IL 1.1V模式	输入漏电流（外部 上拉3 kΩ） VDD = VDD_11_S3	—	5	uA	
CIn 1.1V 模式	输入电容 VDD = VDD_11_S3	—	5	pF	
V _{HYS}	输入迟滞	100	—	mV	
VIH_Min_	输入高电压	0.7×VDD 11 S3_ _	—	V	
VIL_Max	输入低电压	—	0.3×VDD 11 S3_ _		
VSLPRV	接收缓冲器斜率反转	150	—	mV _{PK-PK}	1

备注:

1. 在VIH Min和VIL Max之间发生的斜率反转，在上升或下降边缘上，必须小于VSLPRV（参见图1a）。上升边缘斜率反转，其中信号首先上升到高于VIH Min然后下降到低于VIH Min，必须下降不超过VSLPRV低于VIH Min（参见图1b）。下降边缘斜率反转，其中信号首先下降到低于VIL Max然后上升到高于VIL Max，必须上升不超过VSLPRV高于VIH Min（参见图1c）。

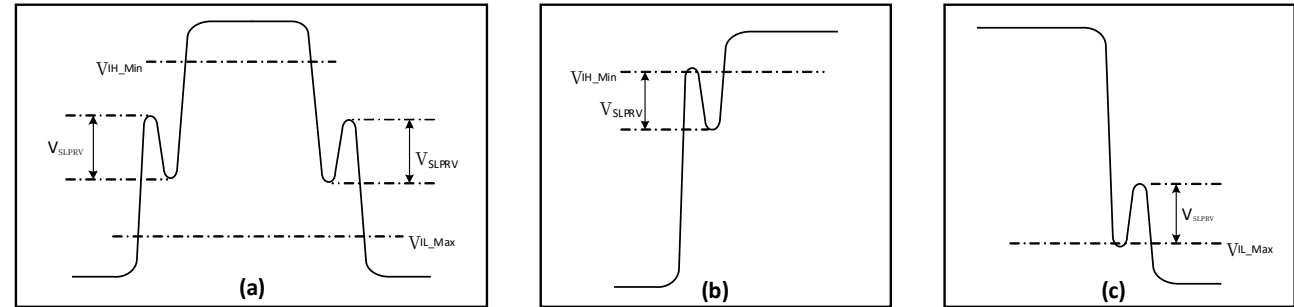


图 1. 斜率反转规范

Table 11. Combined AC and DC Operating Conditions for SEC_SCL and SEC_SDA

Symbol	Parameter	Min	Max	Unit	Notes
V _{IH-DC} 1.1V Mode	Minimum DC voltage at PAD pin that will produce a logic high output on 1.1 V receiver	0.76	—	V	
V _{IH-AC} 1.1V Mode		0.92	—		
V _{IL-DC} 1.1V Mode	Maximum DC voltage at PAD pin that will produce logic low output on 1.1 V receiver	—	0.32		
V _{IL-AC} 1.1V Mode		—	0.095		
V _{OH} 1.1V Mode	Output high voltage (logic 1) @ I = 4 mA	0.75×VDD_11_S3	—		
V _{OL} 1.1V Mode	Output low voltage (logic 0) @ I = 4 mA	—	0.25×VDD_11_S3		
V _{IH-DC} 1.8V Mode	Input high voltage (logic 1)	1.26	1.80		
V _{IH-AC} 1.8V Mode			1.98		1
V _{IL-DC} 1.8V Mode	Input low voltage (logic 0)	0	0.54		
V _{IL-AC} 1.8V Mode		-0.18			1
V _{OH-DC} 1.8V Mode	DC output high voltage (logic 1) @ I = 2 mA	—	—		
V _{OL-DC} 1.8V Mode	DC output low voltage (logic 0) @ I = −2 mA	—	0.36		
I _L 1.1V Mode	Input leakage current (external pull-up of 3 kΩ) VDD = VDD_11_S3	—	5	μA	
I _L 1.8V Mode	Input leakage current	—	10		
C _{In} 1.1V Mode	Input capacitance VDD = VDD_11_S3	—	5	pF	
C _{In} 1.8V Mode	Input capacitance	—	10		
V _{HYS}	Input hysteresis	100	—	mV	

Notes:
1. Input overshoot and undershoot are limited in duration to 25% of the clock period of the highest frequency of the pin's clock.

表 11. SEC_SCL 和 SEC_SDA 的 AC 和 DC 联合工作条件

符号	参数	Min	Max	Unit	备注
VIH-DC 1.1V 模式	PAD 引脚的最小直流电压 将产生一个逻辑高电平 在 1.1 V 接收器上输出	0.76	—	V	
VIH-AC 1.1V 模式		0.92	—		
VIL-DC 1.1V 模式	PAD 引脚的最大直流电压 将产生逻辑低输出 在 1.1 V 接收器上	—	0.32		
VIL-AC 1.1V 模式		—	0.095		
VOH 1.1V 模式	输出高电压（逻辑 1） @ I = 4 mA	0.75×VDD 11 S3_ _	—		
VOL 1.1V 模式	输出低电压（逻辑 0） @ 我 = 4 是A	—	0.25×VDD 11 S3_ _		
VIH-DC 1.8V 模式	输入高电压（逻辑1）	1.26	1.80		
VIH-AC 1.8V 模式			1.98		1
VIL-DC 1.8V 模式	输入低电压（逻辑0）	0	0.54		
VIL-AC 1.8V模式		-0.18			1
VOH-DC 1.8V模式	直流输出高电压（逻辑1） @ I = 2 mA	—	—		
VOL-DC 1.8V模式	DC 输出低电压（逻辑 0） @ I = -2 mA	—	0.36		
IL 1.1V 模式	输入漏电流（外部 上拉 3 kΩ） VDD = VDD 11 S3_ _	—	5	μA	
IL 1.8V 模式	输入漏电流	—	10		
CIn 1.1V 模式	输入电容 VDD = VDD 11 S3_ _	—	5	pF	
CIn 1.8V 模式	输入电容	—	10		
V _{HYS}	输入迟滞	100	—	mV	

备注：
1. 输入过冲和下冲在持续时间上限制为最高频率引脚时钟周期的25%。

Table 12. Combined AC and DC Operating Conditions for Bidirectional Pins:
UART0_CTS_L/UART2_TXD/AGPIO135, UART0_INTR/AGPIO139, UART0_RTS_L/UART2_RXD/AGPIO137, UART0_RXD/AGPIO136, UART0_TXD/AGPIO138, UART1_RXD/AGPIO141, UART1_TXD/AGPIO142, RESET_L, SPI_CS0_L/AGPIO118, SPI_CS1_L/AGPIO119, SPI_CS2_L/AGPIO126, ESPI0_DAT[0]/SPI0_DAT[0]/AGPIO120, ESPI0_DAT[1]/SPI0_DAT[1]/AGPIO121, ESPI0_DAT[2]/SPI0_DAT[2]/AGPIO122, ESPI0_DAT[3]/SPI0_DAT[3]/AGPIO123, ESPI1_DAT[0]/SPI1_DAT[0]/AGPIO131, ESPI1_DAT[1]/SPI1_DAT[1]/AGPIO132, ESPI1_DAT[2]/SPI1_DAT[2]/AGPIO133, ESPI1_DAT[3]/SPI1_DAT[3]/AGPIO134, ESPI_CLK0/SPI_CLK0/AGPIO117, ESPI_CLK1/SPI_CLK1/AGPIO75, ESPI_CLK2/AGPIO74, ESPI_CS0_L/AGPIO124, ESPI_CS1_L/AGPIO125, ESPI_RSTIN_L/KBRST_L/AGPIO129, AGPIO104, AGPIO105, AGPIO106, AGPIO107, GENINT_L/PM_INTR_L/UBM_CPRSNT_CHANGE_DET_L/AGPIO89, SYS_RESET_L/AGPIO1, WAKE_L/AGPIO2, THERMTRIP_L, GENINT_L/PM_INTR_L/UBM_CPRSNT_CHANGE_DET_L/AGPIO89, ESPI_RSTIN_L/KBRST_L/AGPIO129, SGPIO_DATAOUT/AGPIO260, SGPIO_LOAD/AGPIO261, AGPIO256/SGPIO0_CLK, AGPIO257/SGPIO1_CLK/CLK_REQ01_L, AGPIO258/SGPIO2_CLK/CLK_REQ02_L, AGPIO259/SGPIO3_CLK, USB00_OC_L/AGPIO264, USB01_OC_L/AGPIO265, USB10_OC_L/AGPIO16, USB11_OC_L/AGPIO17, PWROK/SVI_RST_L, BP[3:0], XTRIG_L[7:4], SA[1:0], AZ_BITCLK, AZ_RST_L/SW_MDATA1, AZ_SDIN0/SW_MDATA3, AZ_SDIN1/SW_MCLK, AZ_SDIN2/SW_MDATA0, AZ_SDOUT/SW_MDATA2, AZ_SYNC, AGPIO109, AGPIO115/CLK_REQ11_L, AGPIO116/CLK_REQ12_L, AGPIO21, AGPIO22, AGPIO3/SPD_HOST_CTRL_L, AGPIO4/SATA_ACT_L, AGPIO5/DEVSLP0/SATA_ZP0_L, AGPIO6/DEVSLP1/SATA_ZP1_L, AGPIO76/SPI_TPM_CS_L, AGPIO88, DBREQ_L, PCIE_RST0_L/AGPIO266, PCIE_RST1_L/AGPIO26, ESPI0_ALERT_L/AGPIO108, ESPI_RSTOUT_L/AGPIO23, NMI_SYNC_FLOOD_L, PWRGD_OUT/AGPIO12, and RTCCLK

Symbol	Parameter	Min	Max	Unit	Notes
V _{OH-DC}	DC output high voltage (logic 1) @ I = 2 mA	VDD_18 – 0.45	—	V	1
V _{OL-DC}	DC output low voltage (logic 0) @ I = –2 mA	—	0.45		1
V _{IH-DC} V _{IH-AC}	Input high voltage (logic 1)	0.65×VDD_18	VDD_18		2
			1.10×VDD_18		
V _{IL-DC} V _{IL-AC}	Input low voltage (logic 0)	0	0.35×VDD_18	mV	2
		0.10×(–VDD_18)			
V _{hys}	Input hysteresis	150	—		
I _L	Input leakage current	—	5	μA	
C _{In}	Input capacitance	—	1.5	pF	
—	SYS_RESET_L assertion time	10	—	ms	
—	KBRST_L assertion time	0.2	50		

- Notes:
- This parameter does not apply for RESET_L.
 - Input overshoot and undershoot are limited in duration to 25% of the clock period of the highest frequency of the pin's clock.

表 12. 组合交流电和直流电 Operating 双向引脚条件:
UART0_CTS_L/UART2_TXD/AGPIO135, UART0_INTR/AGPIO139, UART0_RTS_L/UART2_RXD/AGPIO137, UART0_RXD/AGPIO136, UART0_TXD/AGPIO138, UART1_RXD/AGPIO141, UART1_TXD/AGPIO142, RESET_L, SPI_CS0_L/AGPIO118, SPI120_CS1_L/AGPIO119, SPI_CS2_L/AGPIO126, ESPI0_DAT[0]/SPI0_DAT[0]/AGPIO ESPI0_DAT[1]/SPI0_DAT[1]/AGPIO121, ESPI0_DAT[2]/SPI0_DAT[2]/AGPIO122, ESPI0_DAT[3]/SPI0_DAT[3]/AGPIO123, ESPI1_DAT[0]/SPI1_DAT[0]/AGPIO131, ESPI1_DAT[1]/SPI1_DAT[1]/AGPIO132, ESPI1_DAT[2]/SPI1_DAT[2]/AGPIO133, ESPI1_DAT[3]/SPI1_DAT[3]/AGPIO134, ESPI_CLK0/SPI_CLK0/AGPIO117, ESPI_CLK1/SPI_CLK1/AGPIO75, ESPI_CLK2/AGPIO74, ESPI_CS0_L/AGPIO124, ESPI_CS1_L/AGPIO125, ESPI_RSTIN_L/KBRST_L/AGPIO129, AGPIO104, AGPIO105, AGPIO106, AGPIO107, GENINT_L/PM_INTR_L/UBM_CPRSNT_CHANGE_DET_L/AGPIO89

SYS_复位_L/AGPIO1, 唤醒_L/AGPIO2, 温度保护_L, GENINT_L/PM_INTR_L/UBM_CPRSNT_CHANGE_DET_L/AGPIO89, ESPI_复位_L/KBRST_L/AGPIO129, SGPIO_数据输出/AGPIO260, SGPIO_LOAD/AGPIO261, AGPIO256/SGPIO0_CLK, AGPIO257/SGPIO1_CLK/CLK_REQ01_L, AGPIO258/SGPIO2_CLK/CLK_REQ02_L, AGPIO259/SGPIO3_CLK, USB00_OC_L/AGPIO264, USB01_OC_L/AGPIO265, USB10_OC_L/AGPIO16, USB11_OC_L/AGPIO17, PWROK/SVI_RST_L, BP[3:0], XTRIG_L[7:4], SA[1:0], AZ_BITCLK, AZ_RST_L/SW_MDATA1, AZ_SDIN0/SW_MDATA3, AZ_SDIN1/SW_MCLK, AZ_SDIN2/SW_MDATA0, AZ_SDOUT/SW_MDATA2, AZ_SYNC, AGPIO109, AGPIO115/CLK_REQ11_L, AGPIO116/CLK_REQ12_L, AGPIO21, AGPIO22, AGPIO3/SPD_HOST_CTRL_L, AGPIO4/SATA_ACT_L, AGPIO5/DEVSLP0/SATA_ZP0_L, AGPIO6/DEVSLP1/SATA_ZP1_L, AGPIO76/SPI_TPM_CS_L, AGPIO88, DBREQ_L, PCIE_RST0_L/AGPIO PCIE_RST1_L/AGPIO26, ESPI0_ALERT_L/AGPIO108, ESPI_RSTOUT_L/AGPIO23, NMI_SYNC_FLOOD_L, PWRGD_OUT/AGPIO12, 和 RTCCLK

符号	参数	Min	Max	Unit	备注
VOH-DC	直流输出高电压（逻辑1） @ I = 2 mA	VDD 18 – 0.45	—	V	1
VOL-DC	DC 输出低电压（逻辑 0） @ I = –2 mA	—	0.45		1
VIH-DC V _{IH-AC}	输入高电压（逻辑 1）	0.65×VDD 18	VDD 18		2
			1.10×VDD _18		
VIL-DC V _{IL-AC}	输入低电压（逻辑0）	0	0.35×VDD 18	mV	2
		0.10×(–VDD_18)			
V _{hys}	输入迟滞	150	—		
I _L	输入漏电流	—	5	μA	
C _{In}	输入电容	—	1.5	pF	
—	SYS_RESET_L 断言时间	10	—	ms	
—	KBRST_L 断言时间	0.2	50		

- 备注:
- 此参数不适用于 RESET_L。
 - 输入过冲和欠冲在持续时间上限制为引脚时钟最高频率的25%。

Chapter 8 Thermal Protection

8.1 THERMTRIP_L Thermal Protection

THERMTRIP functionality is a mechanism to protect the processor from thermal damage. Assertion of THERMTRIP_L indicates the factory determined thermal shutdown temperature ($T_{DIE, SHUTDOWN}$) has been reached or exceeded. The specified power planes must be powered down within a maximum defined time (t_{SD_DELAY}) and the external chipset is expected to place the system in the S5 ACPI state (power off). Deassertion of PWROK resets the THERMTRIP_L assertion.

Additional information in regards to thermal protection circuitry reference designs and thermal solution guidelines can be found in the *Processor Programming Reference (PPR)* documents.

Table 13. Platform Thermal Protection Limits for the Processor

Symbol	Parameter Description	Typ	Max	Unit	Notes
$T_{CTL,SHUTDOWN}$	THERMTRIP shutdown temperature on T_{CTL} scale	115	—	°C	3
$T_{DIE,SHUTDOWN}$	THERMTRIP shutdown die temperature	—	125		1
t_{SD_DELAY}	Maximum allowed time from THERMTRIP_L detection to power plane shutdown to prevent thermal damage to the processor.	—	500	ms	2

- Notes:
- T_{DIE} refers to the physical temperature measured in degrees Celsius. T_{DIE} may not be reported from the processor.
 - Shutdown is specified within this context as ramping down of the power planes to 20% of the normal specified value.
 - A non-slew rate controlled T_{CTL} temperature is used to compare against THERMTRIP shutdown threshold.

第 8 章 热保护

8.1 THERMTRIP L 热保护

THERMTRIP 功能是一种保护处理器免受热损伤的机制。THERMTRIP_L 的断言表示工厂确定的温度关断温度 ($T_{DIE, SHUTDOWN}$) 已达到或超过。指定的电源平面必须在定义的最大时间内 (t_{SD_DELAY}) 关闭，并且外部芯片组预期将系统置于 S5 ACPI 状态（关机）。PWROK 的断言会重置 THERMTRIP_L 的断言。

关于热保护电路参考设计和热解决方案指南的附加信息可以在处理器编程参考（PPR）文档中找到。

表 13. 处理器平台热保护限制

符号	参数描述	Typ	Max	Unit	备注
$T_{CTL,SHUTDOWN}$	THERMTRIP 在 TCTL 尺度上的关断温度	115	—	°C	3
$T_{DIE,关断}$	THERMTRIP 关断芯片温度	—	125		1
$t_{SD_延迟}$	从 THERMTRIP 检测到电源平面关断的最大允许时间以防止过热对处理器的损害。	—	500	ms	2

- 备注:
- T_{DIE} 指的是以摄氏度为单位测量的物理温度。 T_{DIE} 可能不会被处理器报告。
 - 在此上下文中，关机被指定为将电源平面降低到正常指定值的 20%。
 - 使用非斜率控制 T_{CTL} 温度与 *THERMTRIP* 关机阈值进行比较。

Chapter 9 SVI Pins

For implementation details and electrical specifications of the SVC, SVD, SVTI, and SVTO pins, refer to the *AMD Serial VID Interface Version 3 (SVI3) Specification*, publication #56413.

第 9 章 SVI 引脚

有关SVC、SVD、SVTI和SVTO引脚的实现细节和电气规格，请参阅AMD串行VID接口版本3（SVI3）规范，出版物编号#56413。

Chapter 10 SB-TSI Interface

For the non-electrical characteristics of the SB-TSI signals, refer to the “SB Temperature Sensor Interface (SB-TSI)” section in the *Processor Programming Reference (PPR)* documents.

The electrical characteristics of SB-TSI pins are listed in Table 14. Standard mode (100 KHz) and fast mode (400 KHz) are supported; high-speed mode (3.4 MHz) is not supported.

Table 14. Combined AC and DC Operating Conditions for ALERT_L

Symbol	Parameter	Min	Max	Unit
V _{OH-DC}	DC output high voltage (logic 1) @ I = 2 mA	VDD_18_S5 – 0.45	—	V
V _{OL-DC}	DC output low voltage (logic 0) @ I = –2 mA	—	0.45	

Notes:
1. Input overshoot and undershoot are limited in duration to 25% of the clock period of the highest frequency of the pin's clock.

Table 15. Combined AC and DC Operating Conditions for SIC and SID

Symbol	Parameter	Min	Max	Unit
V _{IH-DC 1.1V}	Minimum DC voltage at PAD pin that will produce a logic high output on 1.1 V receiver	0.76	—	V
V _{IH-AC 1.1V}		0.92	—	
V _{IL-DC 1.1V}	Maximum DC voltage at PAD pin that will produce logic low output on 1.1 V receiver	—	0.32	
V _{IL-AC 1.1V}		—	0.095	
I _{L 1.1V}	Input leakage current (external pull-up of 3 kΩ) VDD = VDD_11_S3	—	5	uA
C _{In 1.1V}	Input capacitance VDD = VDD_11_S3	—	5	pF
V _{HYS}	Input hysteresis	100	—	mV
V _{OH 1.1V}	Output high voltage (logic 1) @ I = 4 mA	0.75×VDD_11_S3	—	V
V _{OL 1.1V}	Output low voltage (logic 0) @ I = 4 mA	—	0.25×VDD_11_S3	

第 10 章 SB-TSI 接口

SB-TSI 信号的电气特性，请参阅处理器编程参考 (PPR) 文档中的“SB 温度传感器接口 (SB-TSI)”部分。

SB-TSI 引脚的电气特性列于表 14。支持标准模式 (100 KHz) 和快速模式 (400 KHz)；不支持高速模式 (3.4 MHz)。

表14. ALERT_L的交流 and 直流工作条件组合

符号	参数	Min	Max	Unit
V _{OH-DC}	直流输出高电压（逻辑1） @ I = 2 mA	VDD 18 S5 – 0.45_ _	—	V
V _{OL-DC}	DC输出低电压（逻辑0） @ I = –2 mA	—	0.45	

备注：
1. 在 put overshoot 和 undershoot 的持续时间限制为引脚最高频率时钟周期的 25% 时钟。

表15. SIC和SID的交流直流联合工作条件

符号	参数	Min	Max	Unit
V _{IH-DC 1.1V}	PAD引脚上的最小直流电压 将在...产生一个逻辑高电平输出 1.1 V 接收器	0.76	—	V
V _{IH-AC 1.1V}		0.92	—	
V _{IL-DC 1.1V}	PAD 引脚上的最大直流电压 将在 1.1 V 产生逻辑低输出 接收者	—	0.32	
V _{IL-AC 1.1V}		—	0.095	
I _{L 1.1V}	输入漏电流（外部上拉 3 kΩ） VDD = VDD _11_S3	—	5	uA
C _{In 1.1V}	输入电容 VDD = VDD _11_S3	—	5	pF
V _{HYS}	输入迟滞	100	—	mV
V _{OH 1.1V}	输出高电压（逻辑1） @ I = 4 mA	0.75×VDD 11 S3_ _	—	V
V _{OL 1.1V}	输出低电压（逻辑 0） @ I = 4 mA	—	0.25×VDD 11 S3_ _	

Chapter 11 SATA Interface

For electrical characteristics of the SATA interface signals, refer to the *Serial ATA Revision 3.2 Gold Specification*.

第11章 SATA接口

有关SATA接口信号的电气特性，请参阅Serial ATA Revision 3.2 Gold Specification。

Chapter 12 USB 3/USB 2 Interface

Silicon is USB 3 SuperSpeed compliant.

For electrical characteristics of the USB_SS interface signals, refer to the *USB 3.2 Gen1 Specification*. For electrical characteristics of the USB_HS interface signals, refer to the *USB 2.0 Specification*.

第12章 USB 3/USB 2 接口

硅是 USB 3 SuperSpeed 兼容的。

对于 USB_SS 接口信号的电气特性，请参阅 USB 3.2 Gen1 规范。对于 USB_HS 接口信号的电气特性，请参阅 USB 2.0 规范。

Chapter 13 SPI/eSPI Interface

13.1 SPI

The SPI bus interfaces support single, dual, and quad modes. The SPI interfaces support speeds up to 100 MHz.

Table 16. SPI Timing

Symbol	S-Clock		16 MHz		22 MHz		33 MHz		66 MHz		100 MHz		Notes
	Parameter	Unit	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
t _{SLCH}	SPI_CSx_L or SPI_TPM_CS_L assertion setup	ns	60	—	43	—	28	—	13	—	8	—	1, 2, 3
t _{CHSH}	SPI_CSx_L or SPI_TPM_CS_L assertion hold	ns	60	—	43	—	28	—	13	—	8	—	1, 2, 3
t _{CLQV}	Output valid time	ns	—	3	—	3	—	3	—	3	—	3	1, 2
t _{DS}	Data in setup	ns	3.4	—	3.4	—	3.4	—	3.4	—	3.4	—	1, 2
t _{DH}	Data in hold	ns	1	—	1	—	1	—	1	—	1	—	1, 2
f _R	Clock frequency read instruction	MHz	16.34	17	21.78	23	32.67	34	65.34	67	99	101	
t _{CLH}	Clock high time	ns	26.1	34.5	19.58	25.88	13.0	17.3	6.5	8.6	3.9	6.1	
t _{CLL}	Clock low time	ns	26.1	34.5	19.58	25.88	13.0	17.3	6.5	8.6	3.9	6.1	
t _{CLCH}	Clock rise time	ns	0.1	—	0.1	—	0.1	—	0.1	—	0.1	—	
t _{CHCL}	Clock fall time	ns	0.1	—	0.1	—	0.1	—	0.1	—	0.1	—	

- Notes:
- Trace flight time is not included.
 - The output load is 10 pF.
 - Where x is 1 or 2.

第13章 SPI/eSPI接口

13.1 SPI

SPI 总线接口支持单工、双工和四工模式。SPI 接口支持高达 100 MHz 的速度。

表 16. SPI 定时

符号	S-Clock		16 MHz		22 MHz		33 MHz		66 MHz		100 MHz		备注
	参数	Unit	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
t _{SLCH}	SPI_CSx_L or SPI_TPM_CS_L 断言设置	ns	60	—	43	—	28	—	13	—	8	—	1, 2, 3
t _{CHSH}	SPI_CSx_L or SPI_TPM_CS_L 断言保持	ns	60	—	43	—	28	—	13	—	8	—	1, 2, 3
t _{CLQV}	输出有效时间	ns	—	3	—	3	—	3	—	3	—	3	1, 2
t _{DS}	设置中的数据	ns	3.4	—	3.4	—	3.4	—	3.4	—	3.4	—	1, 2
t _{DH}	保持中的数据	ns	1	—	1	—	1	—	1	—	1	—	1, 2
f _R	时钟频率 读取说明	MHz	16.34	17	21.78	23	32.67	34	65.34	67	99	101	
t _{CLH}	时钟高电平时间	ns	26.1	34.5	19.58	25.88	13.0	17.3	6.5	8.6	3.9	6.1	
t _{CLL}	时钟低电平时间	ns	26.1	34.5	19.58	25.88	13.0	17.3	6.5	8.6	3.9	6.1	
t _{CLCH}	时钟上升时间	ns	0.1	—	0.1	—	0.1	—	0.1	—	0.1	—	
t _{CHCL}	时钟下降时间	ns	0.1	—	0.1	—	0.1	—	0.1	—	0.1	—	

- 备注:
- 飞行时间不包含在内。
 - 输出负载为 10 pF。
 - 其中 x 为 1 或 2。

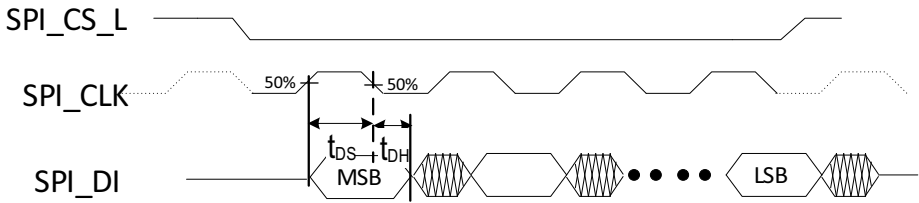


Figure 2. SPI Input Timing

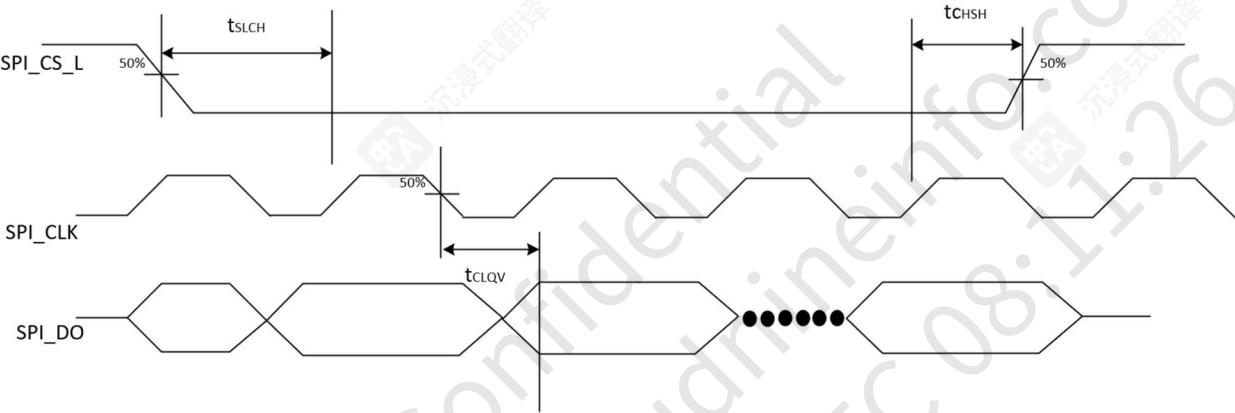


Figure 3. SPI Output Timing

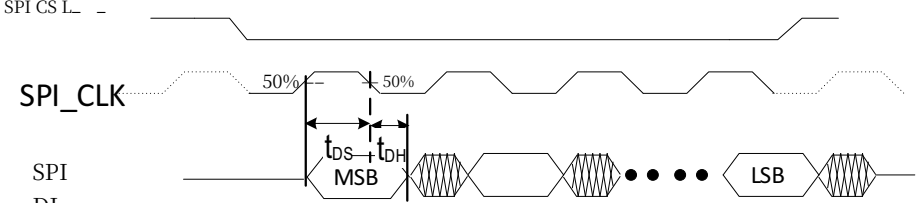


图 2. SPI 输入时序

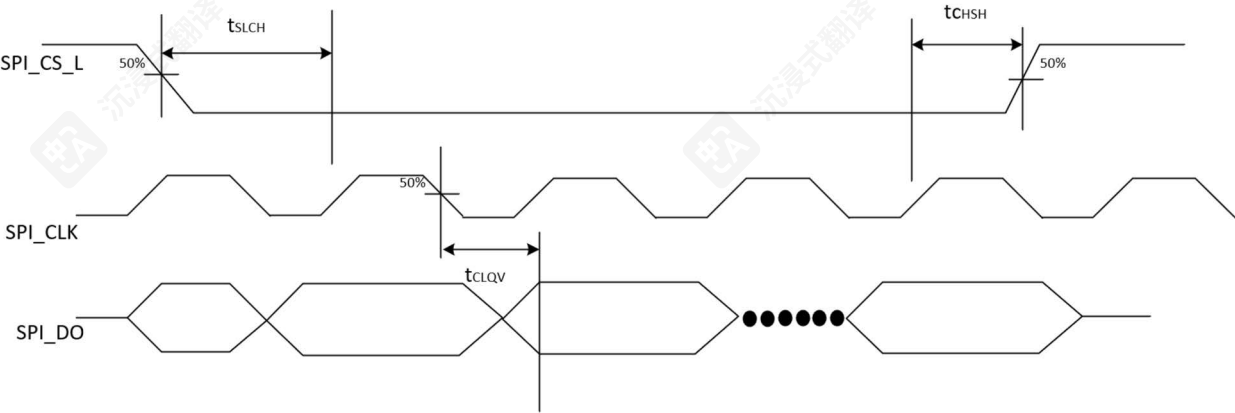


图 3. SPI 输出时序

13.2 eSPI

Refer to the *Interface Base Specification for the Enhanced Serial Peripheral Interface (eSPI), Revision 1.0*.

The eSPI bus interfaces support single, dual, and quad modes. The eSPI bus interfaces support speeds of 16 MHz, 20 MHz, 25 MHz, 33 MHz, 50 MHz, and 66 MHz.

Table 17. eSPI Timing

Symbol	Parameter	Unit	16 MHz		20 MHz		25 MHz		33 MHz		50 MHz		66 MHz	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
T _{CK}	Clock period	ns	60	—	50	—	40	—	30	—	20	—	15	—
T _{CKH}	Clock high time	T _{CK}	0.4	0.6	0.4	0.6	0.4	0.6	0.4	0.6	0.4	0.6	0.4	0.6
T _{CKL}	Clock low time	T _{CK}	0.4	0.6	0.4	0.6	0.4	0.6	0.4	0.6	0.4	0.6	0.4	0.6
T _{CSST}	Chip select setup time	ns	90	—	75	—	60	—	45	—	30	—	22	—
T _{CSHT}	Chip select hold time	ns	60	—	50	—	40	—	30	—	20	—	15	—
T _{CSDT}	Chip select deassertion time	ns	60	—	50	—	40	—	30	—	20	—	15	—
T _{DIST}	Data in setup time	ns	7	—	7	—	7	—	7	—	7	—	7	—
T _{DIHT}	Data in hold time	ns	0.3	—	0.3	—	0.3	—	0.3	—	0.3	—	0.3	—
T _{ODTT}	Output disable time during turnaround	ns	—	18	—	15	—	12	—	9	—	8	—	6
T _{DOVT}	Data output valid time	ns	1	3	1	3	1	3	1	3	1	3	1	3
T _{INIT}	eSPI reset# deassertion to first transaction (GET_CONFIGUR ATION)	us	1	—	1	—	1	—	1	—	1	—	1	—
T _{INIT-FREQ}	Initial bus frequency upon eSPI reset# deassertion	MHz	—	20	—	20	—	20	—	20	—	20	—	20

- Notes:
- All parameters in Table 17 are specified with respect to AMD processor pins and NOT device pins as described in the eSPI Specification.
 - The output load is 10 pF.

13.2 eSPI

请参阅增强型串行外围接口（eSPI）接口基础规范，版本 1.0。

eSPI总线接口支持单、双和四模式。eSPI总线接口支持16 MHz、20 MHz、25 MHz、33 MHz、50 MHz和66 MHz的速度。

Table 17. eSPI Timing

Symbol	参数	Unit	16 MHz		20 MHz		25 MHz		33 MHz		50 MHz		66 MHz	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
T _{CK}	时钟周期	ns	60	—	50	—	40	—	30	—	20	—	15	—
T _{CKH}	时钟高电平时间	T _{CK}	0.4	0.6	0.4	0.6	0.4	0.6	0.4	0.6	0.4	0.6	0.4	0.6
T _{CKL}	时钟低电平时间	T _{CK}	0.4	0.6	0.4	0.6	0.4	0.6	0.4	0.6	0.4	0.6	0.4	0.6
T _{CSST}	芯片选择建立时间 time	ns	90	—	75	—	60	—	45	—	30	—	22	—
T _{CSHT}	Chip select hold time	ns	60	—	50	—	40	—	30	—	20	—	15	—
T _{CSDT}	Chip select deassertion time	ns	60	—	50	—	40	—	30	—	20	—	15	—
T _{DIST}	设置时间中的数据	ns	7	—	7	—	7	—	7	—	7	—	7	—
T _{DIHT}	保持时间中的数据	ns	0.3	—	0.3	—	0.3	—	0.3	—	0.3	—	0.3	—
T _{ODTT}	输出禁用时间 在周转期间	ns	—	18	—	15	—	12	—	9	—	8	—	6
T _{DOVT}	数据输出有效 time	ns	1	3	1	3	1	3	1	3	1	3	1	3
T _{INIT}	eSPI复位# 撤销到第一个 事务 (GET_CONFIGUR ATION)	us	1	—	1	—	1	—	1	—	1	—	1	—
T _{INIT-FREQ}	Initial bus frequency upon eSPI reset# deassertion	MHz	—	20	—	20	—	20	—	20	—	20	—	20

- Notes:
- 表17中所有参数均相对于AMD处理器引脚指定，而非eSPI规范中描述的设备引脚。
- 输出负载为10 pF。

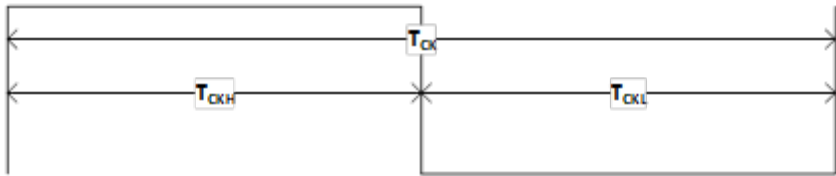


Figure 4. eSPI Clock Timing

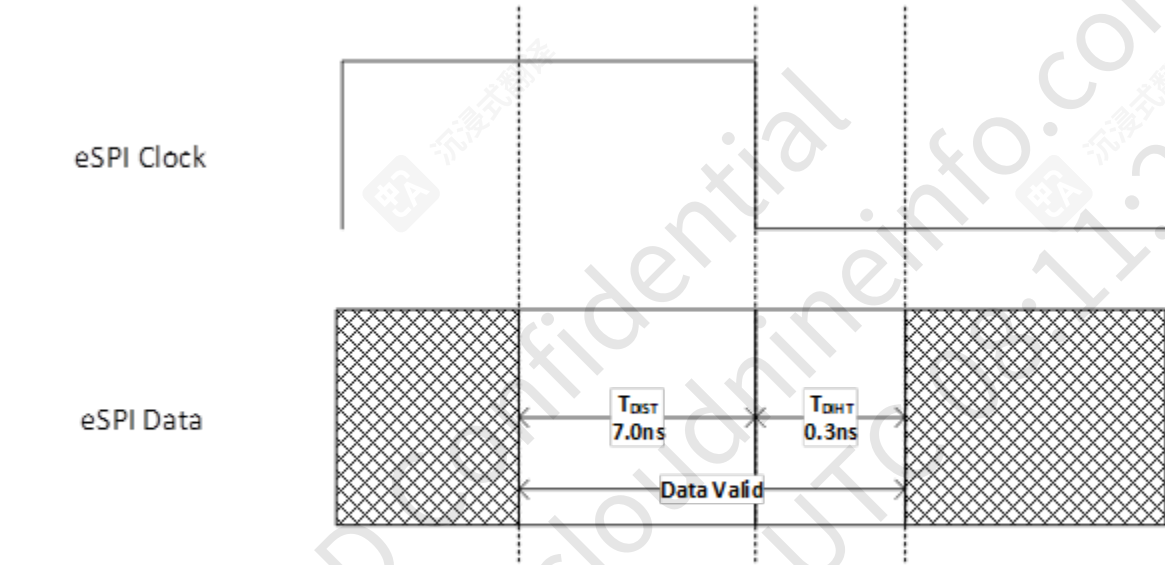


Figure 5. eSPI Input Timing

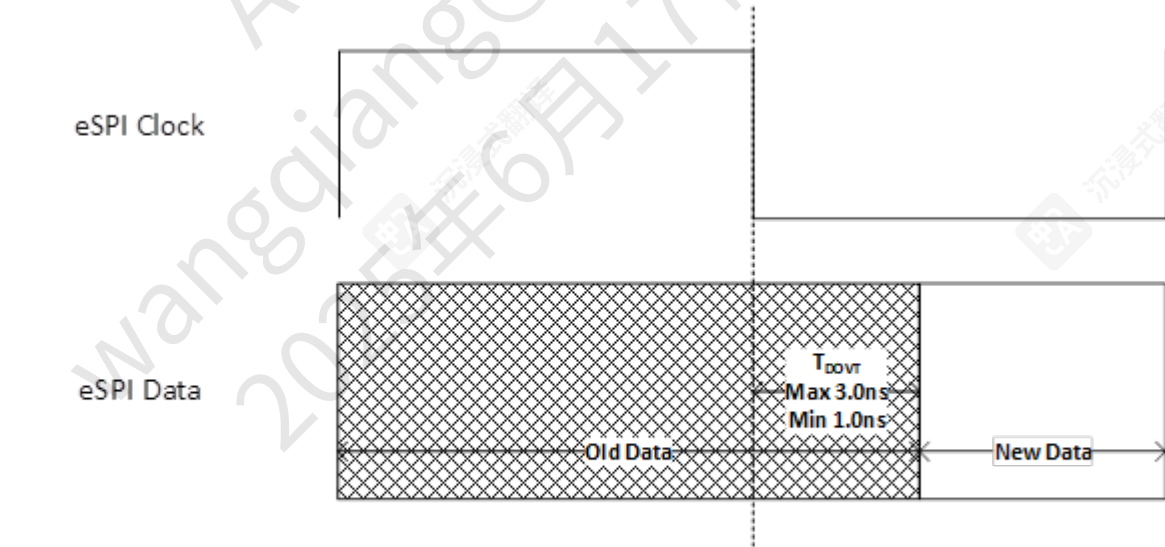


Figure 6. eSPI Output Timing

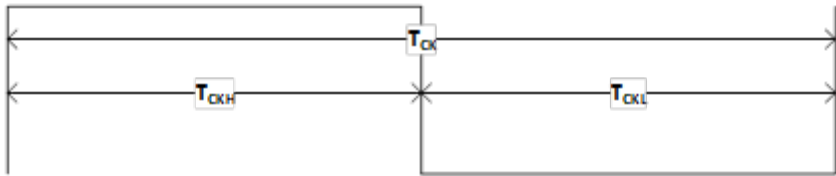


图 4. eSPI 时钟时序

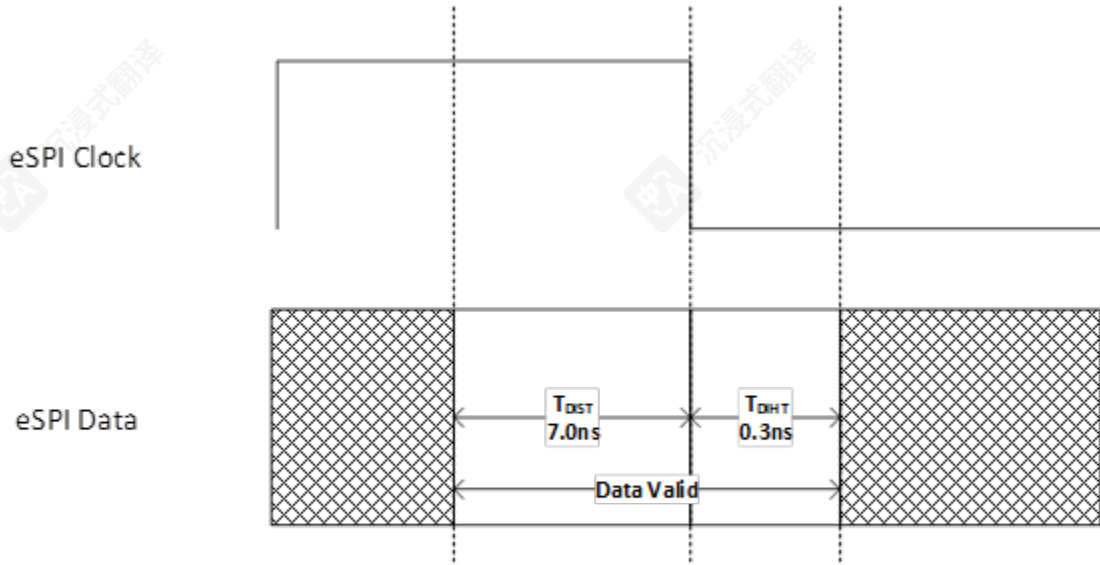


图 5. eSPI 输入时序

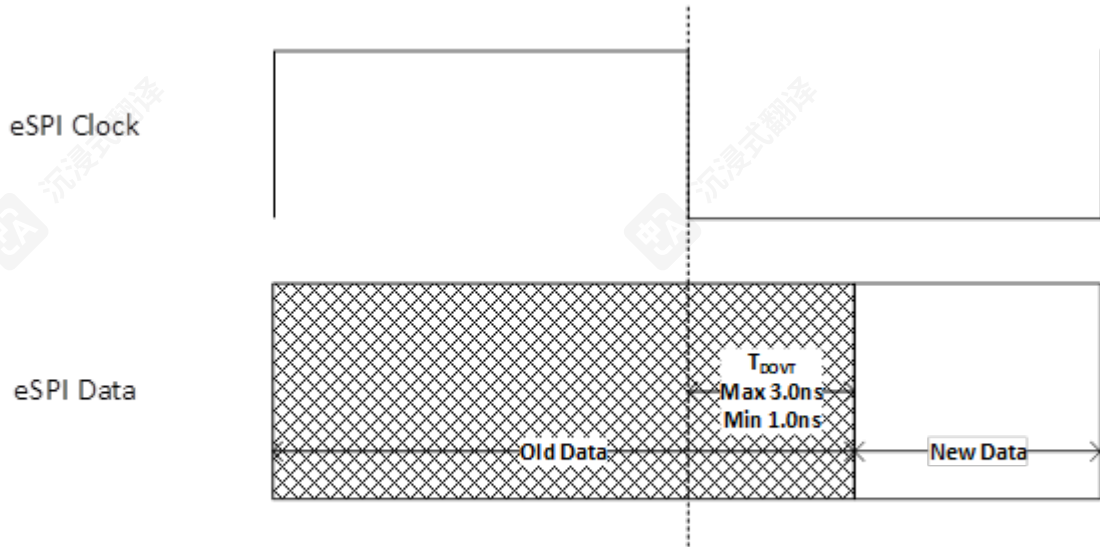


图6. eSPI输出时序

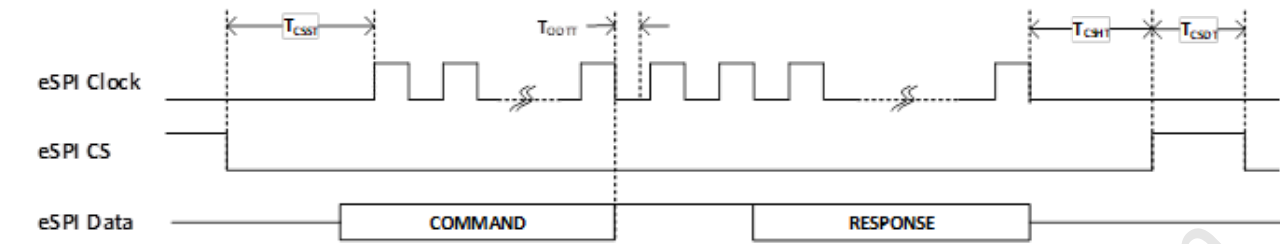


Figure 7. eSPI Protocol Timing

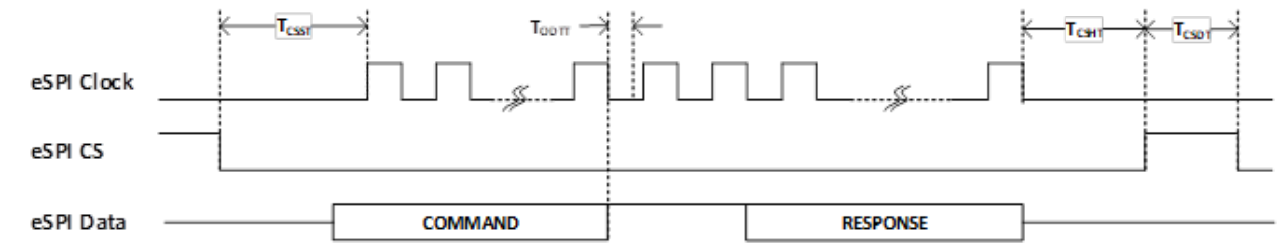


图 7. eSPI 协议时序

Chapter 14 System Management Bus (SMBus) Interface

For electrical characteristics of the SMBus interface signals, refer to the *System Management Bus (SMBus) Specification Version 2.0*.

第 14 章 系统管理总线 (SMBus) 接口

有关SMBus接口信号的电气特性，请参阅系统管理总线（SMBus）规范版本2.0。

Chapter 15 UART Interface

For information on register space, refer to the *Processor Programming Reference (PPR)* documents.

第 15 章 UART 接口

有关寄存器空间的详细信息，请参阅处理器编程参考（PPR）文档。

Chapter 16 I²C/I³C Interface

For information on the I²C and I³C interfaces:

- Refer to the *I²C Specification, I²C Bus Specification and User Manual, UM10204 Rev. 5, 9 October 2012*.
- Refer to the *I³C Basic v1.0 Specification*¹ and Table 10 in this document.

Notes:

- When communicating with I²C legacy devices, *t_{LOW}* can be relaxed to 1200 ns. For details on *t_{LOW}*, refer to Table 57 “I³C Timing Requirements When Communicating with I²C Legacy Devices” in the “I³C Basic v1.0 Specification”.

第 16 章 I²C/I³C 接口

有关I²C和I³C接口的信息：

- 参考 *I²C 规范、I²C 总线规范* 和 *用户手册*，*UM10204* 版本 5，9 2012 年 10 月。
- 参考本文件中的 I³C 基础 v1.0 规范1 和表 10。

注释：

- 在与 I²C 传统设备通信时，*t_{LOW}* 可以放宽到 1200 ns。有关 *t_{LOW}* 的详细信息，请参阅“*I³C 基础 v1.0 规范*”中的表 57 “与 I²C 传统设备通信时的 I³C 时序要求”。

Chapter 17 JTAG Interface

Refer to the *IEEE Standard Test Access Port and Boundary-Scan Architecture Specification, IEEE Std 1149.1tm-2001(R2008)*.

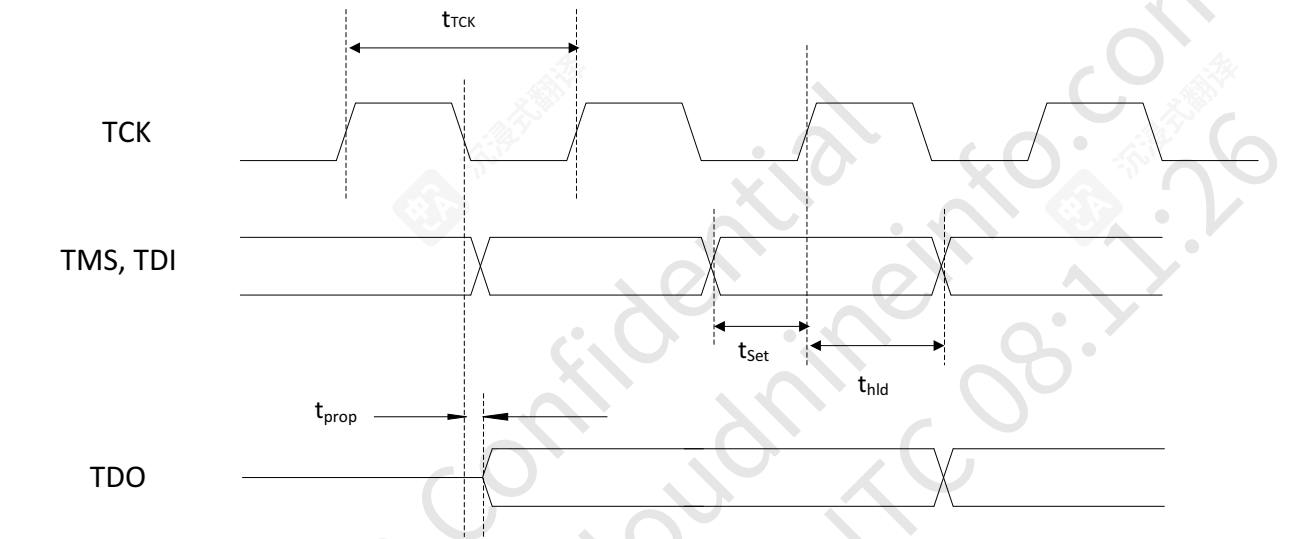


Figure 8. JTAG Timing

Table 18. JTAG Timing Parameters

Parameter	Description	Min	Max	Unit	Notes
t_{TCK}	TCK period	100	1000	ns	1
t_{prop}	TDO output propagation delay from TCK falling edge	—	40		
t_{Set}	TDI, TMS input setup to TCK rising edge	40	—		
t_{hld}	TDI, TMS input hold from TCK rising edge	40	—		

Notes:
1. t_{TCK} may be held low indefinitely.

第 17 章 JTAG 接口

请参阅 *IEEE 标准测试访问端口和边界扫描架构规范, IEEE Std 1149.1tm-2001(R2008)*。

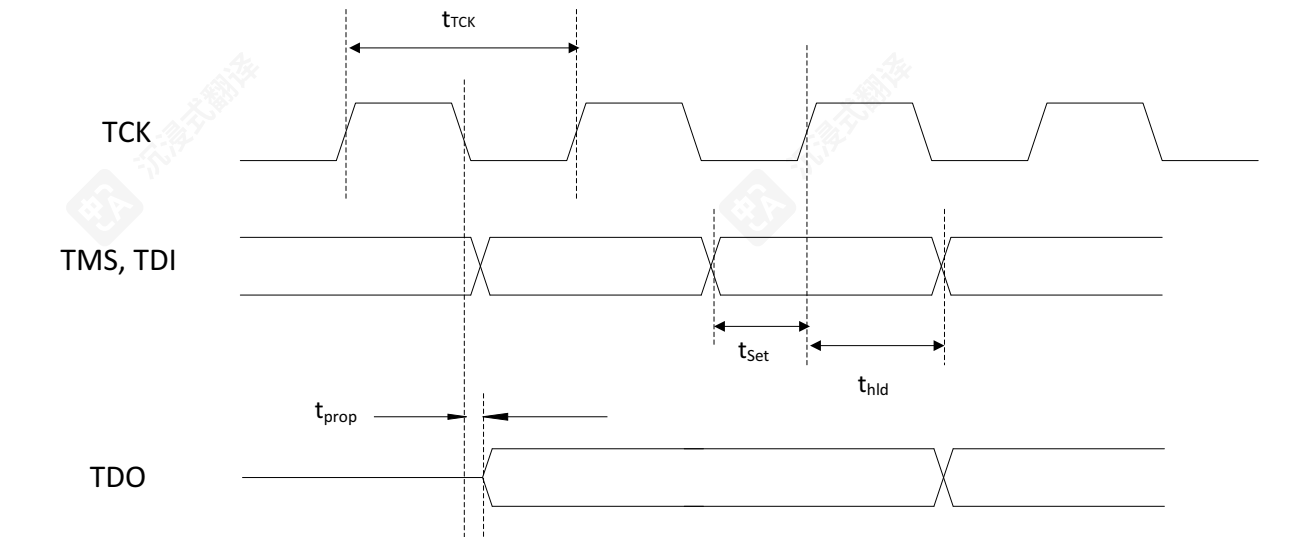


图 8。 JTAG 时序

表 18. JTAG 时序参数

参数	描述	Min	Max	Unit	备注
t_{TCK}	TCK 周期	100	1000	ns	1
t_{prop}	TDO 输出传播延迟从 TCK 下降沿	—	40		
t_{Set}	TDI, TMS 输入建立到 TCK 上升沿	40	—		
t_{hld}	TDI, TMS 输入保持从 TCK 上升 edge	40	—		

备注:
1. t_{TCK} 可持续保持低位。

Chapter 18 Power Supplies and Signal Sequencing

18.1 Power Supply Relationships

Power supply relationships during power-up, power-down, and entry and exit of any power management state must be controlled to ensure proper operation of the device. Power supplies are arranged into power sequencing groups as shown in Table 19 and Table 20. There are no sequencing requirements between power supplies in the same power sequencing group unless specifically noted. ACPI power states S3 and S4 are not supported.

Table 19. Power Sequencing Group Definitions without Coin Cell Battery

Group	System Power Domain	Voltages
Group A	G3/S5	VDDBT_RTC_G, VDD_18_S5, VDD_33_S5, VDDIO_AUDIO
Group B	S3	VDD_11_S3
Group C	S0	VDDIO, VDDCR_SOC, VDDCR_CPU0, VDDCR_CPU1

Table 20. Power Sequencing Group Definitions with Coin Cell Battery

Group	System Power Domain	Voltages
Group A	G3	VDDBT_RTC_G
Group B	S5	VDD_18_S5, VDD_33_S5, VDDIO_AUDIO
Group C	S3	VDD_11_S3
Group D	S0	VDDIO, VDDCR_SOC, VDDCR_CPU0, VDDCR_CPU1

第18章 电源和信号排序

18.1 电源关系

上电、下电以及任何电源管理状态的进入和退出期间，电源之间的关系必须受控以确保设备的正常运行。电源被排列成电源时序组，如表19和表20所示。同一电源时序组内的电源之间没有时序要求，除非特别注明。ACPI电源状态S3和S4不受支持。

表19. 无纽扣电池的电源排序组定义

组	系统电源域	电压
组A	G3/S5	VDDBT_RTC_G, VDD_18_S5, VDD_33_S5, VDDIO_AUDIO
组B	S3	VDD_11_S3
组C	S0	VDDIO, VDDCR_SOC, VDDCR_CPU0, VDDCR_CPU1

表 20. 带纽扣电池的电源时序组定义

组	系统电源域	电压
组A	G3	VDDBT_RTC_G
组B	S5	VDD_18_S5, VDD_33_S5, VDDIO_AUDIO
C组	S3	VDD 11 S3_
D组	S0	VDDIO, VDDCR_SOC, VDDCR_CPU0, VDDCR_CPU1

18.1.1 PWR_GOOD Sequencing Requirements

- All power supplies in all Power Sequencing Groups must be stable and within specification 1 ms before the assertion of PWR_GOOD.
- PWR_GOOD must be deasserted before any supply in any power group is out of specification.

18.1.2 Power-Up Power Sequencing Group Requirements

- All power supplies in Power Sequencing Group A must be stable and within specification before any power supply in Power Sequencing Group B is greater than 10 percent of its specified typical operating value.
- All power supplies in Power Sequencing Group B must be stable and within specification before any power supply in Power Sequencing Group C is greater than 10 percent of its specified typical operating value.
- If using group definitions from Table 20, all power supplies in Power Sequencing Group C must be stable and within specification before any power supply in Power Sequencing Group D is greater than 10 percent of its specified typical operating value.
- All power supplies in Power Sequencing Group C (if using Table 19) or Power Sequencing Group D (if using Table 20) must be stable and within specification 1 ms before the assertion of PWR_GOOD.

18.1.3 Power-Down and Power Failure Power Sequencing Group Requirements

- When SLP_S3_L transitions from 1 to 0, VDDIO, VDDCR_CPU0, VDDCR_CPU1 and VDDCR_SOC must begin to decay.
- When SLP_S5_L transitions from 1 to 0, VDD_11_S3 must begin to decay.
- In an uncontrolled power-down event, when S0 power supplies shut down while S5 supplies are valid, RSMRST_L must be toggled before the system powers on back again to an S0 state.
- No sequencing relationships are required between the Power Sequencing Groups during an uncontrolled power-down or power failure.

18.1.1 PWR_GOOD 顺序要求

- 所有电源序列组中的所有电源必须稳定并在规范1内 ms 在 PWR 断言之前 _好。
- PWR_好必须在任何电源组中的任何电源超出规格之前被取消。

18.1.2 电源上电时序组要求

- 电源时序组A中的所有电源必须在电源时序组B中的任何电源大于其典型工作值的 10%之前稳定且在规格范围内。
- 电源时序组B中的所有电源必须在电源时序组C中的任何电源大于其典型工作值的 10%之前稳定且在规格范围内。
- 如果使用表20中的组定义，电源时序组C中的所有电源必须在电源时序组D中的任何电源大于其典型工作值的10%之前稳定且在规格范围内。
- 所有在电源序列组C（如果使用表19）或电源序列组D（如果使用表20）中的电源必须稳定并在断言PWR前1毫秒内符合规格 _GOOD。

18.1.3 断电和电源故障电源序列组要求

- 当SLP_S3_L从1变为0时，VDDIO、VDDCR_CPU0、VDDCR_CPU1和VDDCR_SOC必须开始衰减。
- 当SLP_S5_L从1变为0时，VDD_11_S3必须开始衰减。
- 在非受控断电事件中，当S0电源关闭而S5电源有效时，RSMRST_L必须在系统重新上电回到S0状态之前切换。
- 在非受控断电或电源故障期间，Power Sequencing Groups 之间无需满足任何排序关系。

18.1.4 Power-Up Power Supply and Signal Sequencing

- PWROK and RESET_L are open-drain signals. PWROK is connected between the sockets in a 2-socket system as shown below. PWROK is also routed to the SVI3 controller. Similarly, RESET_L is connected together across the die in a socket and between the sockets in a 2-socket system.
- After PWROK assertion, Startup-VID values are transmitted on SVI3 interfaces.
- RESET_L remains asserted a minimum of 1 ms after PWROK assertion.
- PWROK distribution skew between the sockets should be less than 8 ns.

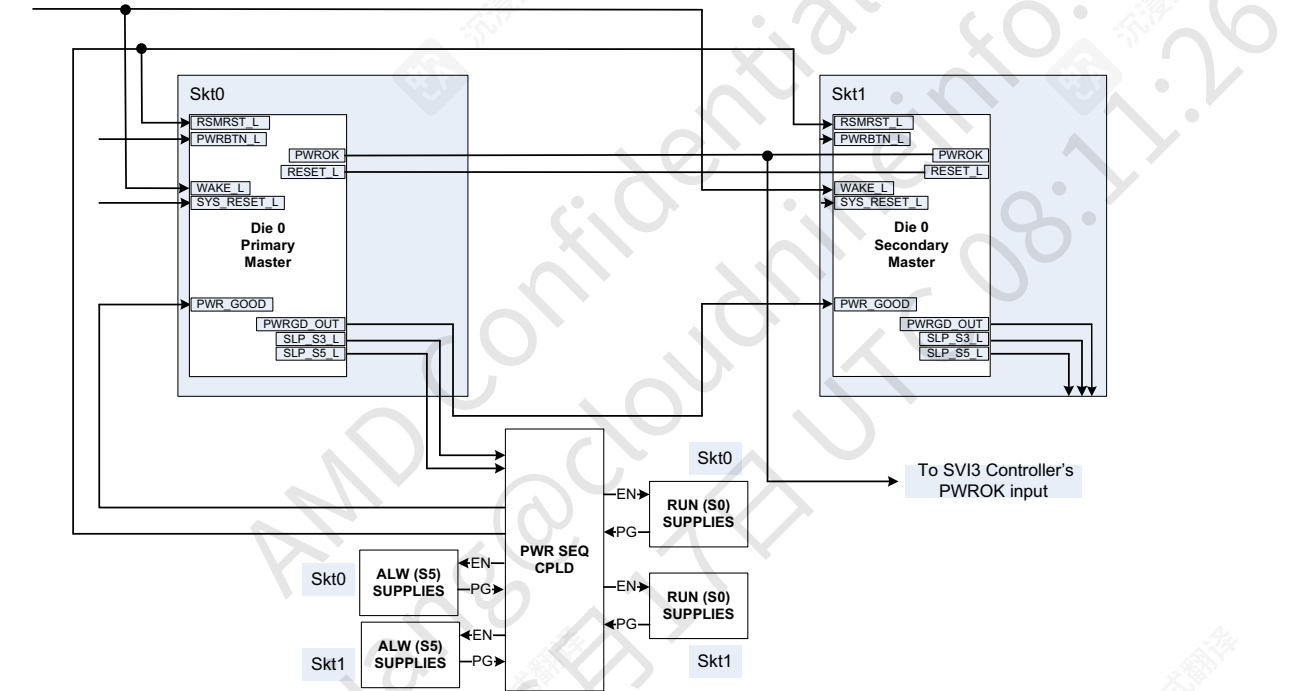


Figure 9. Power and Reset Sequencing Block Diagram

18.1.4 上电电源和信号时序

- PWROK 和 RESET_L 是开漏信号。PWROK 连接到双插槽系统中的插槽之间，如下所示。PWROK 也路由到 SVI3 控制器。类似地，RESET_L 跨越芯片上的插槽连接在一起，并在双插槽系统中连接到插槽之间。
- 在 PWROK 断言后，Startup-VID 值通过 SVI3 接口传输。
- RESET_L 在 PWROK 断言后保持断言的最短时间为 1 ms。
- 插座之间的 PWROK 分配偏差应小于 8 ns。

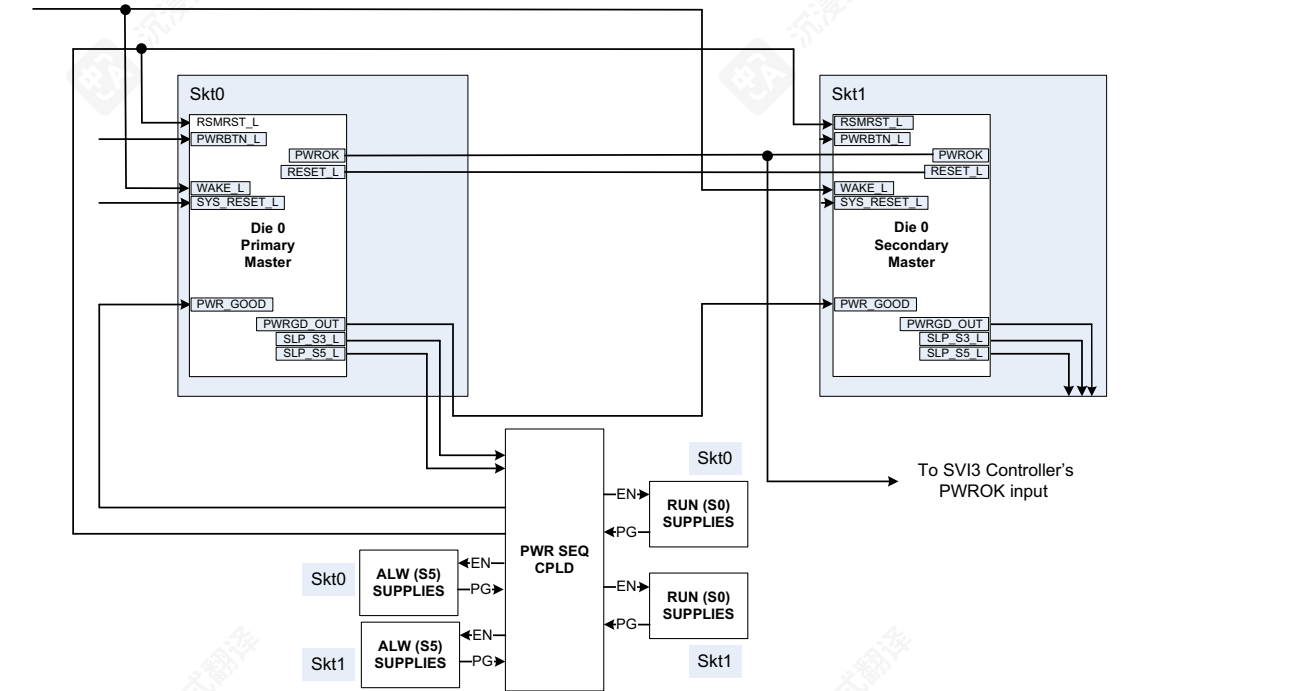


图 9. 电源和复位时序框图

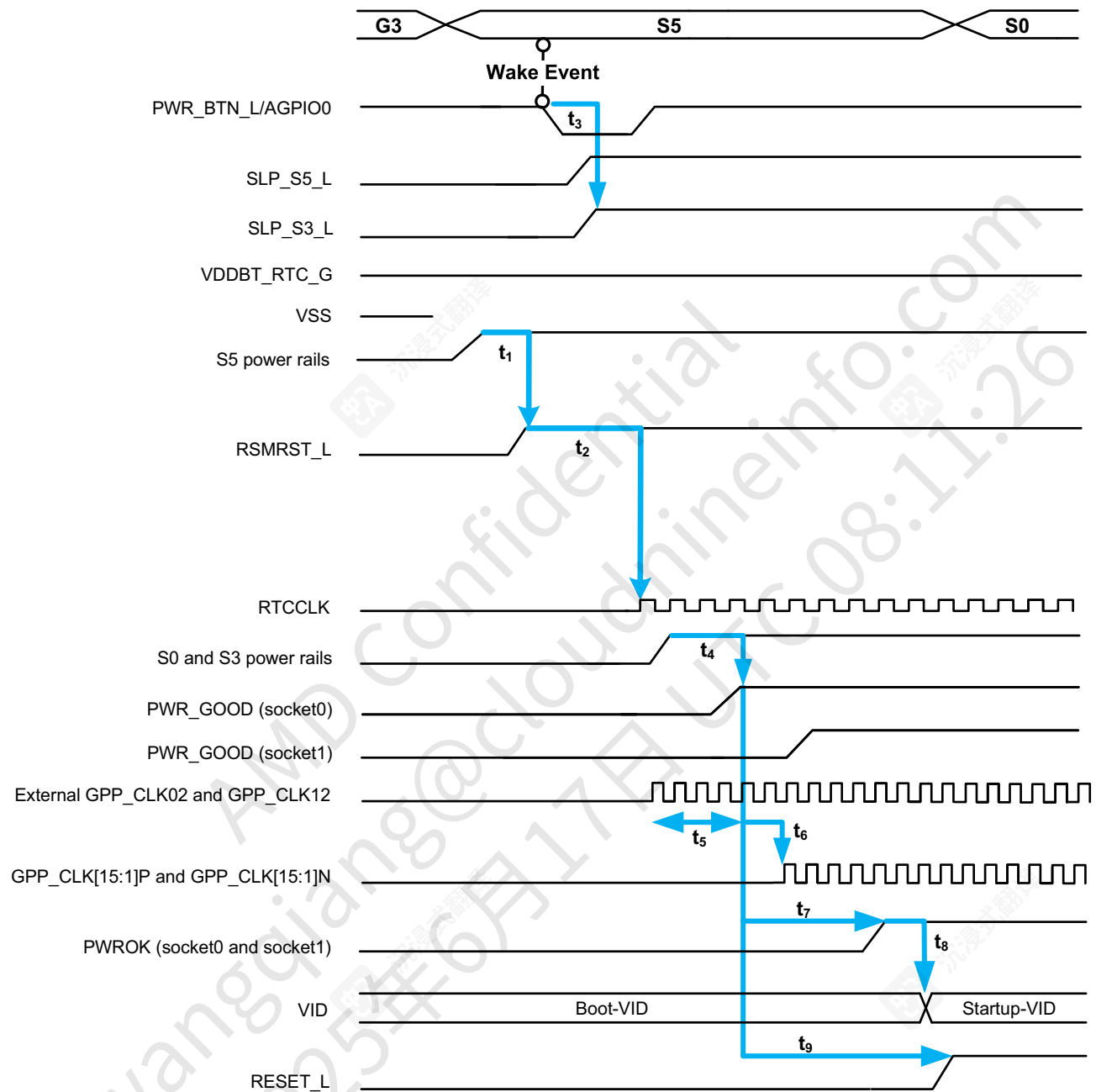


Figure 10. G3->S5->S0 Reset Sequence Timing Diagram

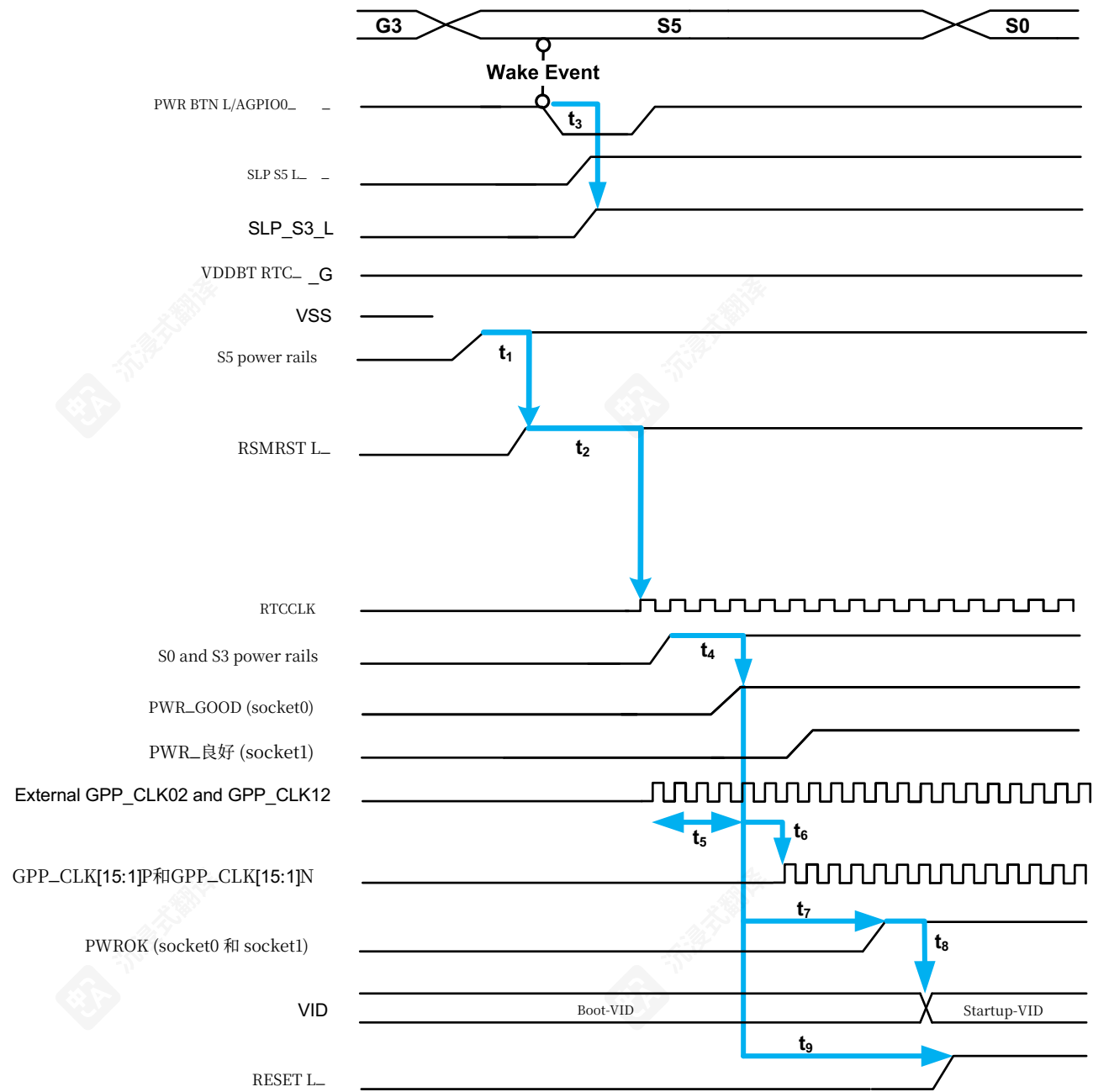


图 10. G3->S5->S0 复位序列时序图

Table 21. G3->S5->S0 Reset Sequencing Timing Parameters

Parameter	Description	Min	Max	Unit
t ₁	From S5 power rails stable to RSMRST_L deassertion.	10	—	ms
t ₂	From RSMRST_L deassertion to RTCCLK assertion.	15	104	
t ₃	From PWR_BTN_L falling to SLP_S3_L deassertion and SLP_S5_L deassertion.	90	360	μs
t ₄	From S0 and S3 power rails stable to PWR_GOOD assertion.	1	—	ms
t ₅	From clock GPP_CLK02 and GPP_CLK12 of both socket0 and socket1 assertion to PWR_GOOD of socket0 assertion.	0	—	ns
t ₆	From PWR_GOOD of socket0 assertion to clock GPP_CLK[15:1]P and GPP_CLK[15:1]N of both socket0 and socket1 assertion.	10.2	26.8	ms
t ₇	1-socket configuration: From PWR_GOOD of socket0 assertion to PWROK assertion.	15.0	20.4	
	2-socket configuration: From PWR_GOOD of socket0 assertion to PWROK assertion.	26.6	36.7	
t ₈	PWROK to Startup-VID delay (if Startup-VID is sent by processor).	200	—	μs
t ₉	1-socket configuration: From PWR_GOOD of socket0 assertion to RESET_L deassertion.	21.2	28.6	ms
	2-socket configuration: From PWR_GOOD of socket0 assertion to RESET_L deassertion.	32	44	

- Notes:
- RESET_L must remain asserted a minimum of 1 ms after PWROK assertion.
 - SYS_RESET_L must remain high a minimum of 1 ms after rising edge of RSMRST_L.
 - Values in this table assume the hardware reset values of the control registers have not been overwritten.

表 21. G3->S5->S0 复位时序参数

参数	描述	Min	Max	Unit
t ₁	从 S5 电源轨稳定到 RSMRST_L 解除。	10	—	ms
t ₂	从 RSMRST_L 解除到 RTCCLK 确认。	15	104	
t ₃	从 PWR_BTN_L 下降到 SLP_S3_L 解除和 SLP_S5_L deassertion。	90	360	μs
t ₄	从 S0 和 S3 电源轨稳定到 PWR_GOOD 断言。	1	—	ms
t ₅	从时钟 GPP_CLK02 和 GPP_socket0 和 socket1 的 CLK12 断言到 PWR_GOOD of socket0 断言。	0	—	ns
t ₆	From PWR_GOOD of socket0 断言到时钟 GPP_CLK[15:1]P and GPP_CLK[15:1]N of both socket0 and socket1 断言。	10.2	26.8	ms
t ₇	1-socket 配置: From PWR_GOOD of socket0 断言到 PWROK 断言。	15.0	20.4	
	2-socket 配置: 从 PWR_GOOD of socket0 断言到 PWROK 断言。	26.6	36.7	
t ₈	PWROK 到 Startup-VID 延迟（如果 Startup-VID 是由处理器发送的）。	200	—	μs
t ₉	1-socket configuration: From PWR GOOD of socket0 assertion to RESET_L deassertion。	21.2	28.6	ms
	2-socket configuration: From PWR GOOD of socket0 assertion to RESET_L deassertion。	32	44	

- Notes:
- RESET_L 必须在 PWROK 断言后保持至少 1 毫秒。
 - SYS_复位_L必须保持高电平至少1 ms，在RSMRST_L上升沿之后。
 - 此表中的值假定控制寄存器的硬件复位值未被覆盖。

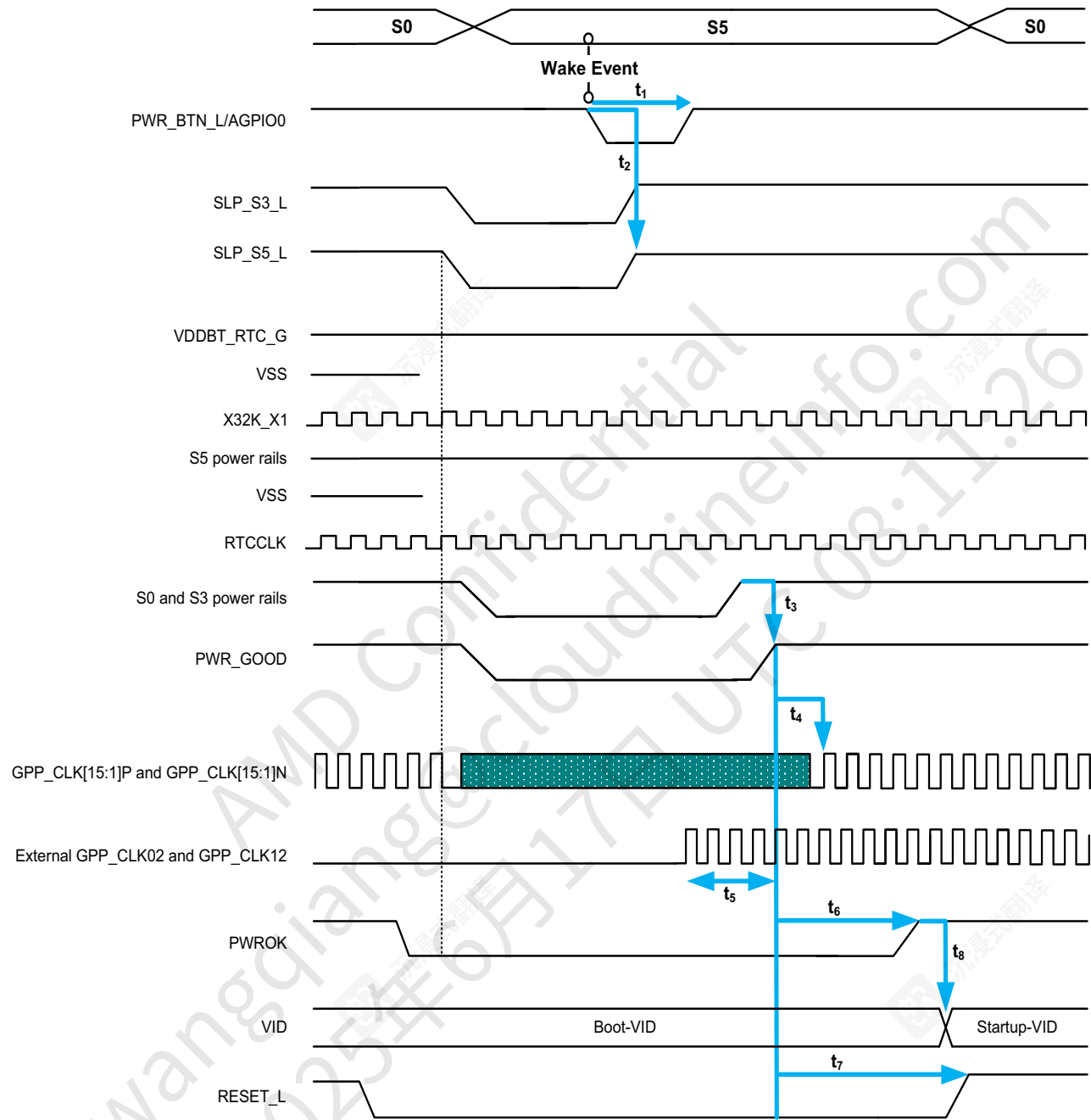


Figure 11. S0->S5->S0 Power Sequencing Timing Diagram

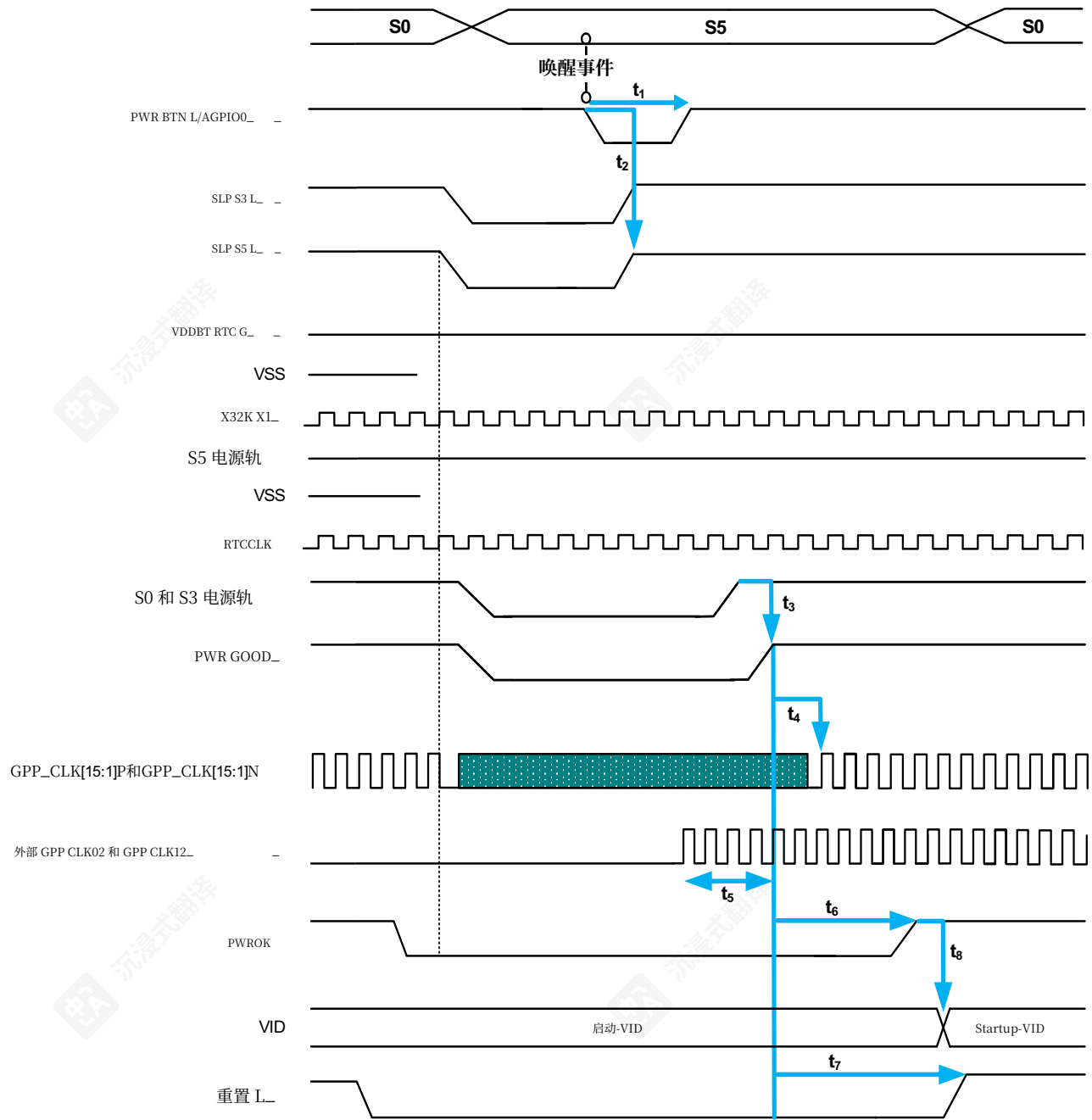


图 11. S0->S5->S0 电源时序时序图

Table 22. S0->S5->S0 Power Sequencing Timing Parameters

Parameter	Description	Min	Max	Unit
t ₁	The minimum assertion time for PWR_BTN_L to wake up the system. Default settings are programmable.	15	—	ms
t ₂	From PWR_BTN_L assertion to SLP_S3_L deassertion and SLP_S5_L deassertion.	14	17	
t ₃	From power rails stable to PWR_GOOD assertion.	1	—	
t ₄	From PWR_GOOD of socket0 assertion to clock GPP_CLK[15:1]P and GPP_CLK[15:1]N of both socket0 and socket1.	10.2	26.8	
t ₅	From clock GPP_CLK02 and GPP_CLK12 of both socket0 and socket1 assertion to PWR_GOOD of socket0 assertion.	0	—	ns
t ₆	1-socket configuration: From PWR_GOOD of socket0 assertion to PWROK assertion.	15.0	20.4	ms
	2-socket configuration: From PWR_GOOD of socket0 assertion to PWROK assertion.	26.6	36.7	
t ₇	1-socket configuration: From PWR_GOOD of socket0 assertion to RESET_L deassertion.	21.2	28.6	
	2-socket configuration: From PWR_GOOD of socket0 assertion to RESET_L deassertion.	32	44	
t ₈	PWROK to Startup-VID delay (if Startup-VID is sent by processor).	200	—	μs

- Notes:
- RESET_L remains asserted a minimum of 1 ms after PWROK assertion.
 - SYS_RESET_L must remain high a minimum of 1 ms after rising edge of RSMRST_L.
 - Values in this table assume the hardware reset values of the control registers have not been overwritten.

表 22. S0->S5->S0 功率时序参数

参数	描述	Min	Max	Unit
t ₁	PWR_BTN_L唤醒系统的最小断言时间。默认设置是可编程的。	15	—	ms
t ₂	从PWR_BTN_L断言到SLP_S3_L取消断言和SLP_S5_L取消断言。	14	17	
t ₃	从电源轨稳定到PWR_GOOD断言。	1	—	
t ₄	从PWR_GOOD of socket0 断言到时钟GPP_CLK[15:1]P 和 GPP_CLK[15:1]N 的 socket0 和 socket1。	10.2	26.8	
t ₅	从时钟 GPP_CLK02 和 GPP_CLK12 的 socket0 和 socket1 断言到 PWR_GOOD of socket0 断言。	0	—	ns
t ₆	1-socket 配置: 从 PWR_GOOD of socket0 断言到 PWROK 断言。	15.0	20.4	ms
	2-socket 配置: 从 PWR_GOOD of socket0 断言到 PWROK 断言。	26.6	36.7	
t ₇	1-socket 配置: 从 PWR_GOOD of socket0 断言到 RESET_L deassertion.	21.2	28.6	
	2-socket configuration: 从 PWR_GOOD of socket0 assertion to RESET_L deassertion.	32	44	
t ₈	PWROK到Startup-VID延迟（如果Startup-VID由处理器发送）。	200	—	μs

- 备注:
- RESET_L 在 PWROK 确认后至少保持 1 ms。
 - SYS_RESET_L 在 RSMRST_L 上升沿后必须保持高电平至少 1 ms。
 - 此表中的值假定控制寄存器的硬件复位值未被覆盖。